

Parcimonious and non-convex optimization applied to statistical learning

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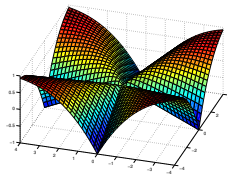
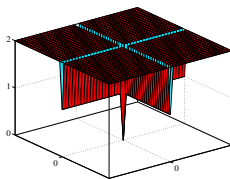
Alain Rakotomamonjy, & Gilles Gasso

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April 10, 2015

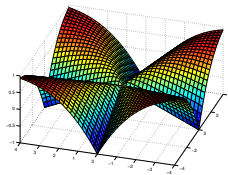
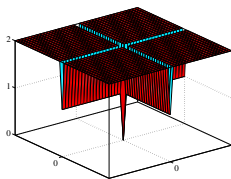
Road map

- 1 Parcimonious and non-convex learning problems
- 2 Mixed integer programming for machine learning
- 3 Continuous relaxations
- 4 Nonsmooth and nonconvex optimization tools
- 5 Optimization algorithm for nonconvex & nondifferential functionals
- 6 A case study: linear regression with MCP



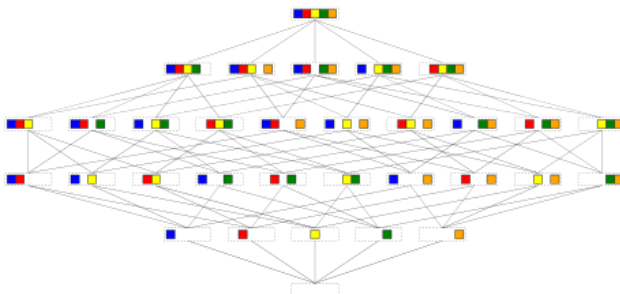
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Variable selection

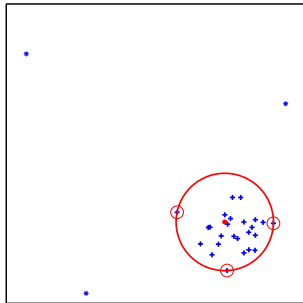
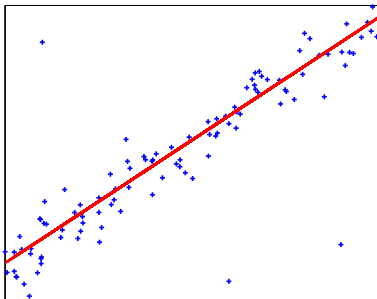
$$f(x_1, \dots, x_j, \dots, x_p) = \sum_{j=1}^{p=5} x_j w_j$$



e.g. gene selection from microarray data, find w sparse such that $Xw = y$

Enumerate of all possible combinations and choose

Outlier detection

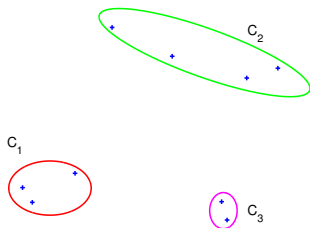


Enumerate of all possible **point configurations** and choose

Clustering

$$Z_{ij} = \begin{cases} 1 & \text{if observation } i \text{ belongs to cluster } j \\ 0 & \text{else} \end{cases}$$

Observation	Cluster 1	Cluster 2	Cluster 3
x_1	1	0	0
x_2	0	1	0
x_3	1	0	0
x_4	0	0	1
x_5	0	1	0
x_6	0	0	1
x_7	0	1	0
x_8	1	0	0
x_9	0	1	0



Enumerate of all possible $(\{0, 1\}, \{0, 1\}, \{0, 1\})^n$ configurations such that each point belongs to only one cluster.

This is a $k = 3$ -partition problem.

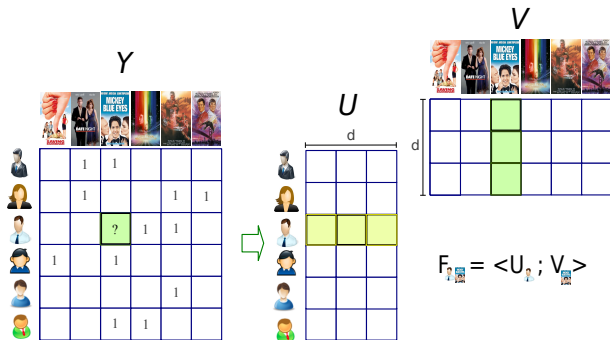
3 d reconstruction problem



Figure : illustration of 3 d reconstruction problem (from University of Tsukuba). Left and middle: two images taken by a stereo camera. Right: True disparity encoded by different gray values which shows the depth of the different objects.

Enumerate of all possible **discrete variations** and choose

Recommendation as a matrix completion problem



$\langle U_i, V_j \rangle$ estimates the *desire* of user i for the product j

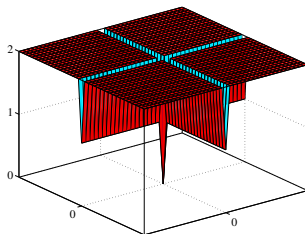
What's common?

Minimize a nonconvex nonsmooth target function

Example (The counting function)

$$\begin{aligned} c : \mathbb{R}^p &\longrightarrow \mathbb{R} \\ w &\longmapsto c(w) = \text{the number of nonzero components } w_i \text{ of } w \end{aligned}$$

It is often called the 0-norm denoted by $c(w) = \|w\|_0$.



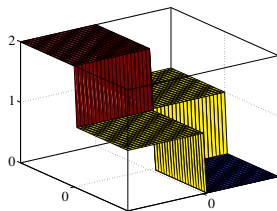
What's common?

Minimize a nonconvex nonsmooth target function

Example (The 0/1 loss function)

$$\begin{aligned} \ell : \mathbb{R}^p &\longrightarrow \mathbb{R} \\ w &\longmapsto \ell(w) = \text{the number of negative components } w_i \text{ of } w \end{aligned}$$

It is related with H the Heavyside step function since $\ell(w) = H(1 - w)$.



What's common?

Minimize a nonconvex nonsmooth target function

Example (The rank function)

$$\begin{array}{ll} r : \mathbb{R}^{n \times p} & \longrightarrow \mathbb{R} \\ A & \longmapsto r(A) = \text{the number of nonzero singular values of } A \end{array}$$

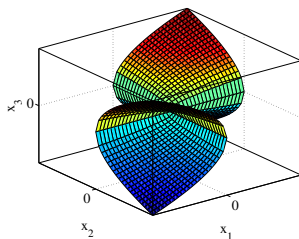


Figure : illustration of the rank 2 manifold of two dimensional symmetric matrices.

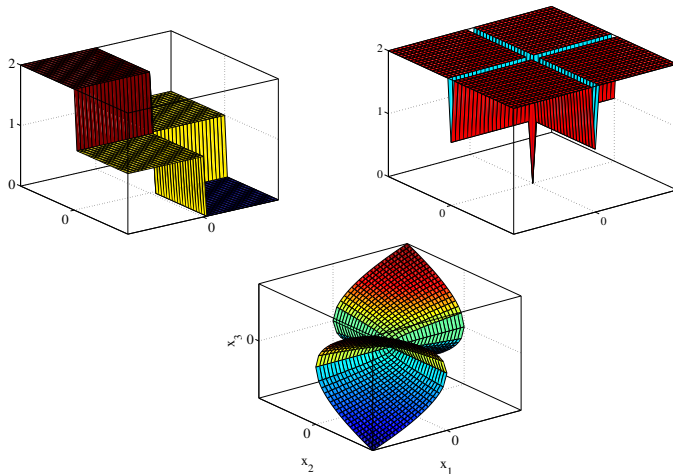


Figure : illustration of $\ell(x)$ the two dimensional 0/1 loss (left), $c(x)$ the counting function (center) and the level one manifold $\{x|r(x) = 1\}$ of the rank function of two dimensional symmetric matrices (right).

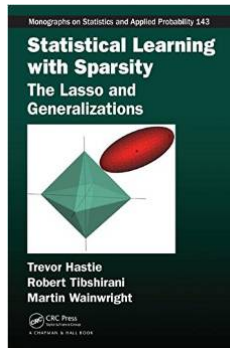
These three functions are nonconvex and nondifferentiable

Why do care about sparsity?

Sparsity is a very useful property of some Machine Learning algorithms.

- Machine Learning is model selection
- **Cheap** to store & transmit
- Sparse coefficients are **meaningful**.
They make more sense.
- More **robust** to errors
- Need **fewer** data to begin with
- provides **scalable** optimization

In the Big Data era, as datasets become larger, it becomes desirable to process the structured information contained within data, rather than data itself.



Nonconvex optimization in machine learning (ML)

Nonconvex Optimization topics OPT2013 workshop Optimization for ML

- Nonsmooth, nonconvex optimization
- Nonconvex quadratic programming, including binary QPs
- Convex Concave Decompositions, D.C. Programming, EM
- Training of deep architectures and large hidden variable models
- Approximation Algorithms

NIPS Workshop on Discrete and Combinatorial Problems in ML since 2009

- Discrete optimization is becoming increasingly important in ML
- Despite discouraging theoretical worst-case results, many practically interesting problems can be much more well behaved
- Submodular functions, the so-called Boolean lattice

Combinatorial problems in machine learning

Many **lattice based** applications

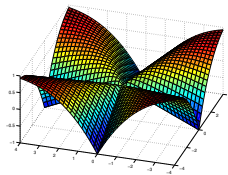
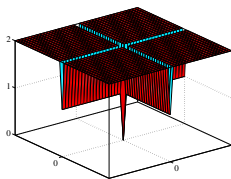
- variable selection, outlier detection
- clustering
- image processing
- compressed sensing, total variations (TV)
- sensor placement
- distribution factorisation
- searching for frequent item sets
- ...

3 ways of solving combinatorial problems

- Fully combinatorial algorithms (greedy search, spanning tree...)
- Mixed integer programming (difficult to scale to large problems)
- Continuous relaxations (not clear when they work)

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Linear system

Given X a $n \times p$ real valued matrix and $y \in \mathbb{R}^n$ a n dimensional vector, we consider the following linear system, that is find $w \in \mathbb{R}^p$ such that

$$Xw = y$$

Given some observed signal y , a measurement matrix (or a dictionary) X , reconstruct w , looking for a sparse solution [see BDE09, for a review].

$n > p$ under determined: no solution (unfeasible) $\rightarrow$ the least square solution
 $w_{ls} = (X^t X)^{-1} X^t y$

$n = p$ a unique solution $w = X^{-1}y$

$n < p$ over determined: no solution $\rightarrow$ choose a sparse one (with n non zero components).

Sparse solutions to linear system

Find a sparse solution to: $Xw = y$

Definition (Sparse vector)

A p dimensional vector $w \in \mathbb{R}^p$ is said to be sparse if $\sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i) = k$ with $k < p$, where $\mathbb{I}$ denotes the indicator function that is

$$\mathbb{I}_{\mathbb{R}^*}(w_i) = \begin{cases} 1 & \text{if } w_i \neq 0 \\ 0 & \text{else.} \end{cases}$$

$$c(w) = \sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i) = \text{the number of nonzero components of } w$$

Problem (The cardinality minimization problem)

$$\begin{cases} \min_{w \in \mathbb{R}^p} & \sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i) \\ \text{with} & Xw = y. \end{cases}$$

The cardinality minimization problem with noise

Relax the constrain by adding some noise.

$$y = Xw + \varepsilon \quad \left\{ \begin{array}{l} \min_{w \in \mathbb{R}^p} \|Xw - y\|_2^2 \\ \min_{w \in \mathbb{R}^p} \sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i). \end{array} \right.$$

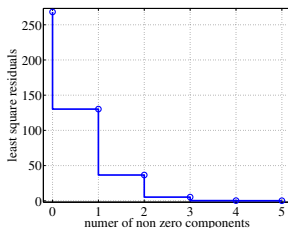


Figure : example of a Pareto's front of the bi objective least square variable selection problem with $n = 10$ and $p = 5$.

Pareto's front

Given two functions L and P from $\mathbb{R}^p$ onto $\mathbb{R}$, a conflicting bi objective optimization problem is

$$\min_{\mathbf{w} \in \mathbb{R}^p} \left\{ \begin{array}{l} L(\mathbf{w}) \\ \text{and} \\ P(\mathbf{w}) \end{array} \right. \quad \text{also denoted} \quad \min_{\mathbf{w} \in \mathbb{R}^p} (L(\mathbf{w}), P(\mathbf{w}))$$

Definition (Pareto dominance)

A vector $\mathbf{w} \in \mathbb{R}^p$ dominates $\mathbf{v} \in \mathbb{R}^p$ when $L(\mathbf{w}) < L(\mathbf{v})$ and $P(\mathbf{w}) < P(\mathbf{v})$.

Definition (Pareto optimal points and Pareto frontier)

A vector $\mathbf{w} \in \mathbb{R}^p$ that can not be dominated (such that $\forall \mathbf{v} \in \mathbb{R}^p$, $\mathbf{v} \neq \mathbf{w}$ we have either $L(\mathbf{w}) < L(\mathbf{v})$ or $P(\mathbf{w}) < P(\mathbf{v})$) is called Pareto optimal. The Pareto frontier (also called Pareto's optimal trade-off curve) of criteria L and P is the set of optimal values $(L(\mathbf{w}), P(\mathbf{w}))$ obtained for all Pareto optimal points $\mathbf{f}$.

3 formulations

Problem (Ivanov regularization)

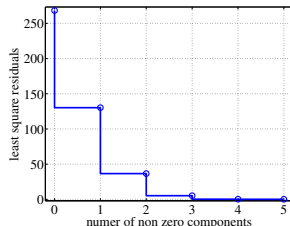
$$w^*(k) \quad \begin{cases} \arg \min_{w \in \mathbb{R}^p} & \|Xw - y\|_2^2 \\ \text{s.t.} & \sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i) \leq k \end{cases}$$

Problem (Phillips regularization)

$$w^*(\tau) \quad \begin{cases} \arg \min_{w \in \mathbb{R}^p} & \sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i) \\ \text{s.t.} & \|Xw - y\|_2^2 \leq \tau \end{cases}$$

Problem (Tikhonov regularization)

$$w^*(\lambda) \quad \arg \min_{w \in \mathbb{R}^p} \quad \|Xw - y\|_2^2 + \lambda \sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i)$$



- The set of all solution is called a regularization path that is

$$\{w_\lambda \in \mathbb{R}^p \mid \lambda > 0\}$$

- The Pareto's front reveals that these three problems don't always provide the same solution.
- These problems are NP-Hard [AK98].

Cardinality constraint as a MIP

Definition (Mixed integer program (MIP))

$$\begin{cases} \min_{w \in \mathbb{R}^p, z \in \mathbb{N}^q} & f(w, z) \\ \text{s.t.} & Aw + Bz = b \\ \text{and} & w, z \geq 0 \end{cases}$$

- Introduce p new binary variable $z \in \{0, 1\}^p$
- For useless variables: $z_j = 0 \Leftrightarrow w_j = 0$

$$\begin{cases} \arg \min_{w \in \mathbb{R}^p} & \|Xw - y\|_2^2 \\ \text{s.t.} & \sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i) \leq k \end{cases} \quad \begin{cases} \min_{w \in \mathbb{R}^p, z \in \{0, 1\}^p} & \frac{1}{2} \|X \text{diag}(z)w - y\|_2^2 \\ \text{s.t.} & \sum_{j=1}^p z_j \leq k \end{cases}$$

Complex non linear interaction between w and z : bad idea

Cardinality constraint as a MIP: good idea

Definition (Mixed integer program (MIP))

$$\begin{cases} \min_{w \in \mathbb{R}^p, z \in \mathbb{N}^q} & f(w, z) \\ \text{s.t.} & Aw + Bz = b \\ \text{and} & w, z \geq 0 \end{cases}$$

Assuming there exist an **upper bound M** for w

$$\sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i) \leq k \quad \Leftrightarrow \quad \begin{cases} z_j \in \{0, 1\}, & j = 1 : p \\ \sum_{i=1}^p z_j \leq k \\ |w_j| \leq z_j M \end{cases}$$

For useless variables: **$z_j = 0 \Leftrightarrow w_j = 0$**

LS with fixed cardinality as a MIQP

$$\left\{ \begin{array}{ll} \arg \min_{w \in \mathbb{R}^p} & \|Xw - y\|_2^2 \\ \text{s.t.} & \sum_{i=1}^p \mathbb{I}_{\mathbb{R}^*}(w_i) \leq k \end{array} \right. \quad \left\{ \begin{array}{ll} \min_{w \in \mathbb{R}^p, z \in \{0,1\}^p} & \frac{1}{2} \|Xw - y\|_2^2 \\ \text{s.t.} & \sum_{j=1}^p z_j \leq k \\ \text{and} & |w_j| \leq z_j M \quad j = 1, p \end{array} \right.$$

Note that the constraint $|w_j| \leq z_j M$ can be replaced by logical implications $z_j = 0 \Rightarrow w_j = 0$. This sort of logical implication can be efficiently handled in a branch-and-bound procedure for MIP problems.

LS with fixed cardinality as a MILP

Definition (mixed integer **linear** program (MILP))

$$\left\{ \begin{array}{ll} \min_{w \in \mathbb{R}^p, z \in \mathbb{N}^q} & f(w, z) = c^t w + d^t z \\ \text{s.t.} & Aw + Bz = b \\ \text{and} & w, z \geq 0 \end{array} \right.$$

When f is quadratic it is called a mixed integer quadratic program

Example (Least absolute residuals with fixed cardinality as a MILP)

$$\left\{ \begin{array}{ll} \min_{w, z \in \{0,1\}^p} & \|Xw - y\|_1 \\ \text{s.t.} & \sum_{j=1}^p z_j \leq k \\ \text{and} & |w_j| \leq z_j M \end{array} \right. \quad \left\{ \begin{array}{ll} \min_{\xi^+, \xi^-, w^+, w^-, z} & c^t(\xi^+ + \xi^-) \\ \text{s.t.} & \xi^+ - \xi^- = X(w^+ - w^-) - y \\ & \sum_{j=1}^p z_j \leq k \\ & w^+ + w^- \leq zM \\ \text{and} & \xi^+, \xi^-, w^+, w^- \geq 0 \end{array} \right.$$

Solving MILP with GLPK

GLPK is a general purpose open source solver¹

```
> help glpk
```

```
[xopt, fmin, status, extra] = glpk(c, a, b, lb, ub, ctype,  
                                vartype, sense, param)
```

Solve an LP/MILP problem using the GNU glpk library. Given three arguments, glpk solves the following standard LP:

min C^*x subject to $A*x \leq b$

but may also solve problems of the form

[**min** | **max**] C^*x

subject to

$A*x$ ["=" | "<=" | ">="] b

$x \geq LB$

$x \leq UB$

¹[glpkmex.sourceforge.net/](http://glpk.mex.sourceforge.net/)

Solving the LA with fixed cardinality using GLPK

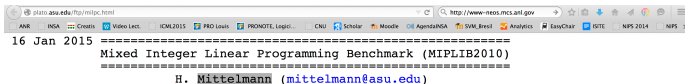
```
n = 10000; p = 5; sig = .2;  
b = [3 2.5 2 1 0]';  
X = randn(n,p);  
y = X*b + sig*randn(n,1);
```

```
In = eye(n); Ip = eye(p); Zp = zeros(n,p);  
c=[ones(2*n,1) ; zeros(3*p,1)];  
A=[-In In X -X Zp; zeros(1,2*(n+p)) ones(1, p) ; Zp' Zp' Ip Ip -Ip];  
k = 3; M = 1; % select k variables  
b=[y;k*M;zeros(p,1)];  
ctype=[repmat('S',n+1,1); repmat('U',p,1)];  
lb=zeros(2*n+3*p,1); ub=[];  
vartype=[repmat('C',2*n+2*p,1); repmat('B',p,1)];  
tic  
[xmin,fmin,status,extra]=glpk(c,A,b,lb,ub,ctype,vartype,1)  
toc  
% n = 10 000, p = 5, k = 3 -> Elapsed time is 9.39 sec.
```

Each year, improvements are expected by a range of 10 to 40%.

Moderate size problems can be solve this way using specific algorithmic framework for convex mixed integer nonlinear programs.

- exact algorithms with global solutions
- available solvers such that such as CPLEX, X-PRESS, GUROBI²...
- on the Mittelman benchmark, commercial solvers such as CPLEX significantly outperforms matlab by a factor of 10 to 100³.



The following codes were run with a limit of 2 hours on the MIPLIB2010 [benchmark set](#) with the [MIPLIB2010](#) scripts (exc Matlab) on two platforms. 1/4 threads: Intel i7-2600, 4 cores, 16GB, 3.4GHz, available memory 12GB 12 threads: Intel Xeon X5680, 12 cores, 32GB, 3.33Ghz, available memory 24GB These are updated and extended versions of the results produced for the MIPLIB2010 [paper](#).

CPLEX-12.6.1: [CPLEX](#)

GUROBI-6.0.0 [GUROBI](#)

ug[SCIP/cpx/spx]-3.1.1: [Parallel development version of SCIP](#) (SCIP+CPLEX/CLP/SOPLEX on 1 thread)

CBC-2.8.10: [CBC](#)

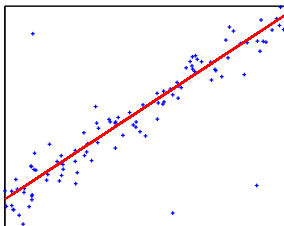
XPRESS-7.8.0: [XPRESS](#)

MATLAB-2014b: [MATLAB](#) (intlinprog; MOSEK only used for input)

²see <http://www.neos-server.org/neos/solvers/index.html#milp>.

³see <http://plato.asu.edu/ftp/milpc.html> for a comparison.

Outlier detection as a MIP



Regression with k outliers.

$$\left\{ \begin{array}{ll} \min_{w \in \mathbb{R}^p, z \in \{0,1\}^n} & \sum_{i=1}^n z_i (X_i^t w - y_i)^2 \\ \text{s.t.} & \sum_{i=1}^n (1 - z_i) \leq k \end{array} \right.$$

This is a mixed 0-1 integer programming problem.

Outlier detection as a MIP

To remove the interaction between z and w , the following linear model is considered

$$y = Xw + \varepsilon + o$$

with k outliers

$$\left\{ \begin{array}{ll} \min_{w \in \mathbb{R}^p, o \in \mathbb{R}^n} & \|Xw + o - y\|^2 \\ \text{s.t.} & \sum_{i=1}^n \mathbb{I}_{\mathbb{R}^*}(o_i) \leq k \end{array} \right\} \quad \left\{ \begin{array}{ll} \min_{w, o, z \in \{0,1\}^n} & \sum_{i=1}^n e_i^2 \\ \text{s.t.} & e - o = Xw + y \\ \text{and} & |o| \leq zM \\ & \sum_{i=1}^n z_i \leq k \end{array} \right.$$

For compressed sensing, Variable selection, outlier detection: mixed 0-1 integer programming problem is ok for moderate size problems.

It is NP hard but It is improving.

3 d reconstruction from stereo images

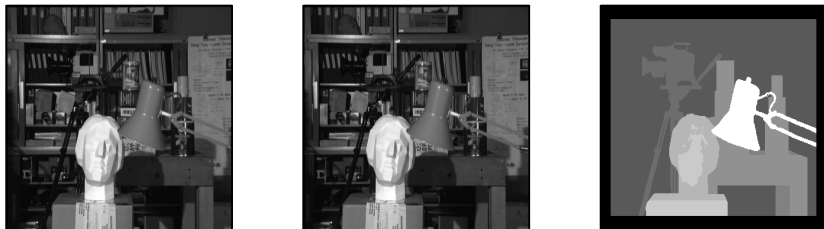


Figure : illustration of 3 d reconstruction problem (from University of Tsukuba). Left and middle: two images taken by a stereo camera. Right: True disparity encoded by different gray values which shows the depth of the different objects.

The classical (discrete) Potts model is, for a given vector $y \in \mathbb{R}^p$

$$\begin{cases} \min_{w \in \mathbb{R}^p} \frac{1}{2} \|y - w\|^2 \\ \min_{w \in \mathbb{R}^p} \sum_{i=1}^{p-1} \mathbb{I}_{\{w_i \neq w_{i+1}\}} \end{cases}$$

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3 formulations

$$\begin{cases} \min_{w \in \mathbb{R}^p} \frac{1}{2} \|y - w\|^2 \\ \text{with } \sum_{i=1}^{p-1} \mathbb{I}_{\{w_i \neq w_{i+1}\}} \leq k \end{cases} \quad \begin{cases} \min_{w \in \mathbb{R}^p} \sum_{i=1}^{p-1} \mathbb{I}_{\{w_i \neq w_{i+1}\}} \\ \text{with } \frac{1}{2} \|y - w\|^2 \leq \tau \end{cases}$$

$$\min_{w \in \mathbb{R}^p} \frac{1}{2} \|y - w\|^2 + \lambda \sum_{i=1}^{p-1} \mathbb{I}_{\{w_i \neq w_{i+1}\}}$$

Potts prior [Pot52]

$$\begin{cases} \min_{w \in \mathbb{R}^p} & \frac{1}{2} \|y - w\|^2 \\ \text{s.t.} & \sum_{i=1}^{p-1} \mathbb{I}_{\{w_i \neq w_{i+1}\}} \leq k \end{cases}$$

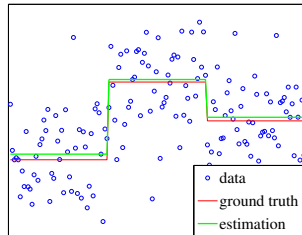
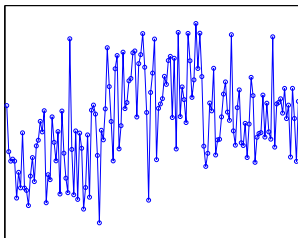


Figure : illustration of the MILP solution of the Potts model with $n = 150$ computed with matlab and glpk (elapsed time is 0.085 sec.).

Total variations

Example (LA with fixed number of discontinuities as a MILP)

$$\left\{ \begin{array}{ll} \min_{w, z \in \{0,1\}^p} & \|w - y\|_1 \\ \text{s.t.} & \sum_{j=1}^{p-1} z_j \leq k \\ \text{and} & |w_j - w_{j+1}| \leq z_j M \\ & j = 1, p-1 \end{array} \right. \quad \left\{ \begin{array}{ll} \min_{w, \xi^+, \xi^-, d^+, d^-, z} & c^t(\xi^+ + \xi^-) \\ \text{s.t.} & \xi^+ - \xi^- = w - y \\ \text{and} & \xi^+, \xi^-, d^+, d^- \geq 0 \\ & \sum_{j=1}^{p-1} z_j \leq k \\ & Cw = d^+ - d^- \\ & d^+ + d^- \leq zM \end{array} \right.$$

This is a MILP (mixed integer linear program).

```

n = 150;
y = [randn(n/3,1) ; randn(n/3,1)+2 ; randn(n/3,1)+1];
yt = [zeros(n/3,1) ; ones(n/3,1)*2 ; ones(n/3,1)*1];

c=[ zeros(n,1) ; ones(2*n,1) ; zeros(3*n-3,1)];
In = eye(n); Inm1 = eye(n-1);
C = In - diag(ones(n-1,1),1); C(end,:) = [];
M = 2;
A=[ In -In In zeros(n,3*n-3) ; C zeros(n-1,2*n) -Inm1 Inm1 zeros(n-1) ; zeros
    (1,5*n-2) ones(1,n-1) ; zeros(n-1,3*n) Inm1 Inm1 -M*Inm1];
k = 2;
b=[y ; zeros(n-1,1) ; k ; zeros(n-1,1)];
ctype=[repmat('S',2*n,1); repmat('U',n-1,1)];
lb=[-inf*ones(n,1) ; zeros(5*n-3,1)]; ub=[];
vartype=[repmat('C',5*n-2,1); repmat('B',n-1,1)];
par.msglev=3;

tic
[xmin,fmin,status,ext]=glpk(c,A,b,lb,ub,ctype,vartype,1,par)
toc

% > Elapsed time is 0.084581 seconds.

```

The elapsed time is sensitive to the choice of M .

Single facility location with outliers

In 1857 J.J. Sylvester posed the problem: It is required to find the least circle which shall contain a given set of points in the plane.

$$\begin{cases} \min_{R, c \in \mathbb{R}^p} R \\ \text{with } \|x_i - c\|^2 \leq R, \quad i = 1, \dots, n \end{cases} \quad (1)$$

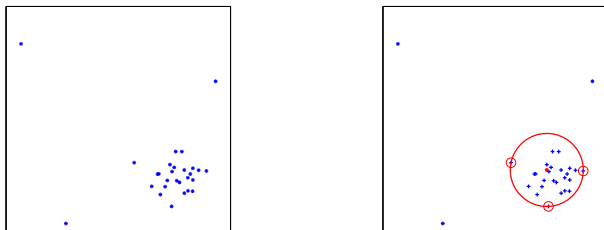


Figure : SVDD with outliers

SVDD with outliers

k outliers treated with specific slack variables ξ

$$\xi = \begin{cases} 0 & \text{if } x_i \text{ is not an outlier} \\ \|x_i - c\|^2 - R & \text{if } x_i \text{ is an outlier} \end{cases}$$

$$\left\{ \begin{array}{ll} \min_{R, c \in \mathbb{R}^p, \xi \in \mathbb{R}^n} & R \\ \text{with} & \|x_i - c\|^2 \leq R + \xi_i, \quad i = 1, \dots, n \\ \text{and} & \xi_i \geq 0, \quad i = 1, \dots, n \\ \text{and} & \sum_{i=1}^n \mathbb{I}_{\mathbb{R}^*}(\xi_i) = k \end{array} \right. \quad (2)$$

Multiple location facility - Clustering

Maximum integer programming (MIP)-Diameter with K cluster

$$\left\{ \begin{array}{ll} \min_{R_k, z_{ik} \in \{0,1\}} & \max(R_1, \dots, R_k, \dots, R_K) \\ \text{with} & z_{ik} z_{jk} \|x_i - x_j\|^2 \leq R_k, \quad i, j = 1, \dots, n \\ \text{and} & \sum_{k=1}^K z_{ik} = 1 \quad i = 1, \dots, n \end{array} \right.$$

As it is, the MIP-Diameter model is a nonconvex bilinear mixed-integer programming model
a MIQP formulation

$$\left\{ \begin{array}{ll} \min_{R_k, c_k, z_{ik} \in \{0,1\}} & \max(R_1, \dots, R_k, \dots, R_K) \\ \text{with} & \|x_i - c_k\|^2 \leq R_k + (1 - z_{ik})M, \quad k = 1, \dots, K \\ \text{and} & \sum_{k=1}^K z_{ik} = 1 \quad i = 1, \dots, n \end{array} \right.$$

that can be reduced to a MILP

Classification and the indicator function

In the two class problem, the cost of making decision d while the true label is y is 0/1 loss given by

$$\ell_{0/1}(d, y) = \begin{cases} 1 & \text{if } d \neq y \\ 0 & \text{else} \end{cases}$$

Assuming $y \in \{-1, 1\}$ and the decision being the sign of some f so that $d = \text{sgn}(f)$, the 0/1 loss can be seen as an indicator function

$$\ell_{0/1}(f, y) = \mathbb{I}_{\mathbb{R}^-}(yf).$$

Empirical loss minimization

$$\min_f \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\mathbb{R}^-}(y_i f(x_i))$$

The hinge loss can be seen its ℓ_1 relaxation

$$\min_f \frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i f(x_i))$$

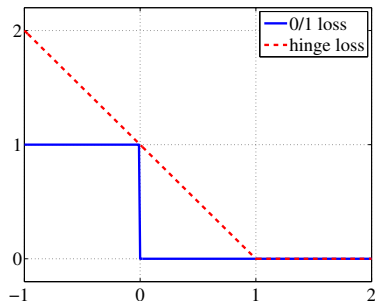


Figure : 0/1 loss classification loss $\mathbb{I}_{\{\mathbb{R}^-\}}(u)$ and the hinge loss $\max(0, 1 - u)$.

SVM and feature selection [WEST03]

$$\left\{ \begin{array}{l} \min_{w,b} \quad \sum_{i=1}^p \mathbb{I}_{\{w_i \neq 0\}} \\ \text{with} \quad y_i(w^t x_i + b) \geq 1 \end{array} \right.$$

outlier detection: remove k outliers

$$\left\{ \begin{array}{l} \min_{w,b} \quad \sum_{i=1}^p \mathbb{I}_{\{w_i \neq 0\}} \\ \text{with} \quad y_i(w^t x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, i = 1, n \\ \sum_{i=1}^n \mathbb{I}_{\{\xi_i \neq 0\}} \leq k \end{array} \right.$$

Matrix completion

Netflix matrix completion, low rank matrix completion, covariance matrix estimation (Recover Inverse Covariance Matrix, sparse + low rank), image approximation, machine learning, control and computer vision (see [Le13]).

Assume the $n \times p$ matrix Y is observed.

$$Y = X^* + \mathcal{E}$$

Low rank hypothesis

$\exists 0 < k < \min(n, p)$, U a $n \times k$ matrix, and V a $p \times k$ matrix such that $X = UV^t$

The rank of a matrix is the corresponding to the number of linearly independent rows or columns of the matrix,

The low rank estimation of X^* based on Y arise in different application fields. Again a bi objective optimization problem

$$\begin{cases} \min_{U,V} & \|Y - UV^t\|^2 \\ \text{s.t.} & \text{rank}(U, V) \leq k \end{cases}$$

low rank matrix completion, M observed values the incomplete frobenius norm.

$$\begin{cases} \min_{U,V} & \sum_{m=1}^M (Y_{i(m),j(m)} - U_{i(m),\bullet} V_{j(m),\bullet}^t)^2 \\ \text{s.t.} & \text{rank}(U, V) \leq k \end{cases}$$

Rank Minimization Problem (RMP) and the rank constrained minimization (RCM) Problems

$$\left\{ \begin{array}{ll} \min_{U,V} & \text{rank}(UV^t) \\ \text{s.t.} & \sum_{m=1}^M (Y_{ij(m)} - U_{i(m),\bullet} V_{j(m),\bullet}^t)^2 \leq \tau \end{array} \right. \quad \left\{ \begin{array}{ll} \min_{U,V} & \sum_{m=1}^M (Y_{ij(m)} - U_{i(m),\bullet} V_{j(m),\bullet}^t)^2 \\ \text{s.t.} & \text{rank}(U, V) \leq k \end{array} \right.$$

$$\min_{U,V} \sum_{m=1}^M (Y_{i(m),j(m)} - U_{i(m),\bullet} V_{j(m),\bullet}^t)^2 + \lambda \text{rank}(U, V)$$

The only (useful) topological property of the rank function is that it is lower-semicontinuous.

Robust PCA: a multi objective optimization problem

$$\left\{ \begin{array}{ll} \min_{U,V,S} & \text{rank}(UV^t) \\ \min_{U,V,S} & \|S\|_0 \\ \text{s.t.} & \|Y - UV^t - S\|^2 \leq \tau \end{array} \right.$$

Machine learning as a bi objective minimization problem

$$\left\{ \begin{array}{ll} \min_{f \in H} & \text{fitting the data} \\ \min_{f \in H} & \text{regular solution} \end{array} \right.$$

Model $H + \text{loss} + \text{penalty}$

- nonconvex models H
 - ▶ Deep neural networks (Yann's tutorial)
 - ▶ discriminative HMM
- Nonconvex nondiscriminative loss 0/1 loss, outlier detection
- Nonconvex nondiscriminative penalty compressed sensing, variable selection, rank minimization

Fitting the data

$$\begin{aligned}\ell: \quad \mathcal{H}, \mathcal{X}, \mathcal{Y} &\longrightarrow \mathbb{R}^+ \\ f, w, y &\longmapsto \ell(f, w, y)\end{aligned}$$

	differentiable	singular
convex	logistic $\log(1 + \exp^{-yf(w)})$ L2-gaussian $(f(w) - y)^2$	hinge $(yf(w) - 1)_+$ L1-laplacian $ f(w) - y $
nonconvex	sigmoid $1 - \tanh(yf(w))$ Cauchy $\log(1 + (f(w) - y)^2)$	$\ell_{0/1}$ $ \text{sign}(f(w)) - y $ Lp - square root $ f(w) - y ^{1/2}$

Table : some example of fitting criteria ℓ . Cost reported line *convex* are convex while the one of the line below are not. Regarding the hinge loss, the following notations are used $(x)_+ = \max(x, 0)$. for classification, label are coded $y \in \{-1, 1\}$.

$$L(f, S_n) = \sum_{i=1}^n \ell(f, w_i, y_i)$$

Select a regular hypothesis

$$\begin{aligned} P : \mathcal{H} &\longrightarrow \mathbb{R}^+ \\ f &\longmapsto P(f) \end{aligned}$$

	differentiable	singular
convex	$\mathcal{L}_2$ $\ f\ _2^2$	$\mathcal{L}_1$ $\ f\ _1$
nonconvex	normalized $\sum_j \frac{\alpha_j^2}{1 + \alpha_j^2}$	Counting $c(w) \mathcal{L}_q, q < 1$ $\ f\ ^q$

Table : examples of regularity criteria P . For the normalized one, hypothesis are supposed to have the following form $f(w) = \sum \alpha_j \phi_j(w)$ for a certain given dictionary of function $\{\phi_j\}_{j=1,p}$.

To formulate these criteria the hypothesis set $\mathcal{H}$ is assumed to be a normed vector space. Their precise definition depends on the nature of the hypothesis set.

Antoniadis' penalties

Table 1 Examples of convex penalty functions

Convex	
Smooth at zero	Singular at zero
1. $\psi(\beta) = \beta ^\alpha, \alpha > 1$	6. $\psi(\beta) = \beta \quad \psi'(0^+) = 1$
2. $\psi(\beta) = \sqrt{\alpha + \beta^2}$	7. $\psi(\beta) = \alpha^2 - (\beta - \alpha)^2 I\{ \beta < \alpha\} \quad \psi'(0^+) = 2\alpha$
3. $\psi(\beta) = \log(\cosh(\alpha\beta))$	
4. $\psi(\beta) = \beta^2 - (\beta - \alpha)^2 I\{ \beta > \alpha\}$	
5. $\psi(\beta) = 1 + \beta /\alpha - \log(1 + \beta /\alpha)$	

Table 2 Examples of non-convex penalty functions

Non-convex	
Smooth at zero	Singular at zero
8. $\psi(\beta) = \alpha\beta^2/(1 + \alpha\beta^2)$	12. $\psi(\beta) = \beta ^\alpha, \alpha \in (0, 1) \quad \psi'(0^+) = \infty$
9. $\psi(\beta) = \min\{\alpha\beta^2, 1\}$	13. $\psi(\beta) = \alpha \beta /(1 + \alpha \beta) \quad \psi'(0^+) = \alpha$
10. $\psi(\beta) = 1 - \exp(-\alpha\beta^2)$	14. $\psi(0) = 0, \psi(\beta) = 1, \forall \beta \neq 0 \quad \text{discont.}$
11. $\psi(\beta) = -\log(\exp(-\alpha\beta^2) + 1)$	15. $\psi(\beta) = \log(\alpha \beta + 1) \quad \psi'(0^+) = \alpha$
	16. $\int_0^\beta \psi'(u)du \quad \psi'(\beta) = \alpha\{I\{ \beta \leq \alpha\} + \frac{(a\alpha - \beta)_+}{(a-1)\alpha} I\{ \beta > \alpha\}\},$ $a > 2$

Figure : classification of some penalty functions according to [AGN11].

non-convex non-differentiable problems are of two kinds

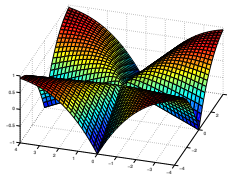
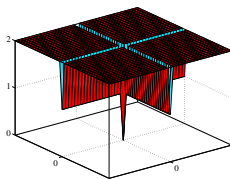
mixed integer programming
continuous programming

- For moderate size mixed integer programming is ok,
 - ▶ After some reformulation to get standard MILP or QP
 - ▶ You've got a global solution.
 - ▶ Develop the proper theory for the three functions such as *discrete convex analysis* (summodularity, $M^\natural$ convexity and associated duality) and associated algorithms. (cf K. Murota's tutorial at NIPS Workshop on Discrete and Combinatorial Problems in Machine Learning (DISCML) 2013: Theory and Applications⁴).
- For larger problem: use continuous relaxation to get efficiency
 - ▶ Relaxation may provide better generalization
 - ▶ But a local minima (local vs. global)

⁴misojiro.t.u-tokyo.ac.jp/~murota/paper/131209NIPS-DISCML.pdf

Road map

- 1 Parcimonious and non-convex learning problems
- 2 Mixed integer programming for machine learning
- 3 Continuous relaxations**
- 4 Nonsmooth and nonconvex optimization tools
- 5 Optimization algorithm for nonconvex & nondifferential functionals
- 6 A case study: linear regression with MCP



The need for continuous relaxation

- Mathematical reason:
The objective is discontinuous
- Computational reason:
The problem is known to be NP-hard
- Statistical reason:
From a machine learning perspective, as we already mentioned, we are often more interested in the generalization performance.
A lot of variance remains because the discontinuities.

Convex continuous relaxations: the L2 relaxation

$c(w)$ = the number of nonzero components

$\ell(e)$ = the number of negative components

$r(A)$ = the number of nonzero singular values

$$\|w\|_2^2 = \sum_{j=1}^p w_j^2$$

$$\|e\|_2^2 = \sum_{i=1}^n \max(e_i, 0)^2$$

$$\|A\|_2^2 = \sum_{j=1}^p \sigma_j^2$$

Example : the ridge regression, L2 SVM

Problems

- No real equivalence
- No sparsity: doesn't scale

Convex relaxations; The L1 relaxation

$c(w)$ = the number of nonzero components

$\ell(e)$ = the number of negative components

$r(A)$ = the nb of nonzero singular values

$$\|w\|_1 = \sum_{j=1}^p |w_j|$$

$$\|e\|_1 = \sum_{i=1}^n \max(e_i, 0)$$

$$\|A\|_* = \text{tr} \left(\sqrt{A^t A} \right) = \sum_{j=1}^p \sigma_j$$

Example : the Lasso regression, compressed sensing, C-SVM, nuclear norm minimization

Advantages

- in some conditions it provides the same solution
- Fast and scalable algorithms

Convex relaxations: the adaptive L1 relaxation

for a given set of coefficients β

$c(w) =$ the number of nonzero components

$\ell(e) =$ the number of negative components

$r(A) =$ the nb of nonzero singular values

$$\|w\|_1 = \sum_{j=1}^p \beta_j |w_j|$$

$$\|e\|_1 = \sum_{i=1}^n \beta_i \max(e_i, 0)$$

$$\|A\|_* = \text{tr} \left(\sqrt{A^t A} \right) = \sum_{j=1}^p \beta_j \sigma_j$$

How to choose the β ?

take the L2 relaxation: back to these problems...

Convex relaxations: problems with the L1 relaxation

Problems with L1 relaxation

- Scaling/Shrinkage Problem:
the L1 bias, the L1 norm may over-shrink large magnitude coefficients.
- Correlated variables problem: no more L0 – L1 equivalence
- Extra dictionary parameters may be hard to estimate: Blind Image Deblurring (degenerate solution is favored)

Non convex continuous relaxations

$\ w\ _0$	???	$\ w\ _1$	$\ w\ _2$
<i>sparse</i>	<i>sparse</i>	<i>sparse</i>	<i>NON</i>
<i>NON</i>	<i>NON</i>	<i>convex</i>	<i>convex</i>

Three strategies for sparse continuous relaxation:

- relax the 0–1 variables: $z_j \in \{0, 1\} \longrightarrow z_j \in [0, 1]$ continuous
- relax the functions: $c(w) \longrightarrow \tilde{c}(w)$ continuous
- project a L2 solution:

Continuous variable relaxation

- start with the MIP formulation (with binary variables)
- replace $z_j \in \{0, 1\}$ by a continuous one $z_j \in [0, 1]$
- favoring the extreme values of the interval, for instance by imposing $z_j(1 - z_j) = 0$

Example (Best subset selection in regression [TTA10])

$$\left\{ \begin{array}{ll} \min_{w \in \mathbb{R}^p, z \in \{0,1\}^p} & \frac{1}{2} \|Xw - y\|_2^2 \\ \text{s.t.} & \sum_{i=1}^p z_j \leq k \\ \text{and} & |w_j| \leq z_j M \end{array} \right. \quad \left\{ \begin{array}{ll} \min_{w \in \mathbb{R}^p, z \in [0,1]^p} & \frac{1}{2} \|Xw - y\|_2^2 \\ \text{s.t.} & \sum_{i=1}^p z_j \leq k \\ \text{and} & |w_j| \leq z_j M \\ \text{and} & z_j(1 - z_j) = 0, \quad j = 1, p \end{array} \right.$$

Best subset selection in regression (II)

Remove the non linear constraints

The relaxed penalized version of this problem is, for a given $\rho > 0$ and considering $\sum_{j=1}^p z_j(1 - z_j)$

$$\left\{ \begin{array}{ll} \min_{w \in \mathbb{R}^p, z \in [0,1]^p} & \frac{1}{2} \|Xw - y\|_2^2 - \frac{\rho}{2} \|z\|_2^2 \\ \text{s.t.} & \sum_{j=1}^p z_j = k \\ \text{and} & |w_j| \leq z_j M \end{array} \right.$$

Best subset selection in regression (III)

Another continuous relaxation

At the optimum

$$\begin{cases} w_j^* = 0 \Rightarrow z_j^* = 0 \\ w_j^* \neq 0 \Rightarrow z_j^* = 1 \end{cases} \quad \Rightarrow \quad \sum_{i=1}^p w_j^* (1 - z_j^*) = 0.$$

so that minimizing $\sum_{i=1}^p |w_j| (1 - z_j)$ would produce the desired effect.

$$\begin{cases} \min_{w \in \mathbb{R}^p, z \in [0,1]^p} & \frac{1}{2} \|Xw - y\|_2^2 + \tau \sum_{i=1}^p |w_j| (1 - z_j) \\ \text{s.t.} & \sum_{i=1}^p z_j \leq k. \end{cases}$$

with no more need of the upperbound M . For well chosen values of τ this problem is equivalent to the least square minimization with cardinality constraint [TTA10] Proposition 7.3.1.2

More relaxations

More relaxations can be founded in [SZL13]. Note that another analogous formulation is close to the non negative garrote [Bre95]

$$\left\{ \begin{array}{ll} \min_{w, z \in \{0,1\}^p} & \frac{1}{2} \|X \text{diag}(z)w - y\|_2^2 \\ \text{s.t.} & \sum_{i=1}^p z_j \leq k \end{array} \right.$$
$$\left\{ \begin{array}{ll} \min_{w, z \in \mathbb{R}^p} & \frac{1}{2} \|X \text{diag}(z)w - y\|_2^2 \\ \text{s.t.} & \sum_{i=1}^p z_j \leq k \\ \text{and} & 0 \leq z_j, \quad i = 1, p \end{array} \right.$$

Non convex approximation of the counting function c

Lq Bridge: the q norm [FF93]

$$\tilde{c}(w) = \|w\|_q^q = \sum_{i=1}^p |w_i|^q$$

Non convex for $0 < q < 1$ and

$$\lim_{q \rightarrow 0} \|w\|_q = c(w)$$

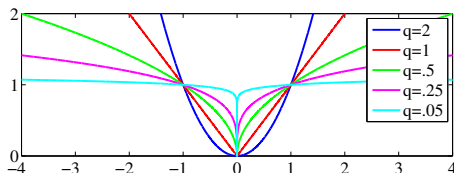


Figure : Lq-norm functions bridges L1 and L0 (the counting function c).

Non convex approximation of the counting function c

Exponential approximation [BMR98]

$$e_b(w) = \sum_{i=1}^p \left(1 - \exp^{-\frac{|w_i|}{b}} \right)$$

$$\lim_{b \rightarrow 0} e_b(w) = c(w)$$

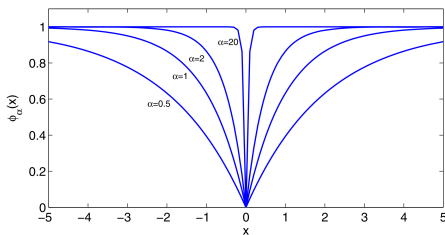


Figure : Exponential approximation functions with $\alpha = 1/b$ (from [SZL13]).

Non convex approximation of the counting function c

Log-sum approximation [WEST03]

$$l_b(x) = \frac{1}{\log(1 + 1/b)} \sum_{i=1}^p \log \left(1 + \frac{|w_i|}{b} \right)$$

$$\lim_{b \rightarrow 0} l_b(w) = c(w)$$

The $\log \ell_1 / \ell_2$ ratio regularization function [RPD⁺14]

$$\varphi_b(x) = \log \left(b + \sum_{i=1}^p |w_i| \right) - \log \|w\|_2$$

with $\|w\|_2^2 = \sum_{i=1}^p w_i^2$.

More approximations

Geman Penalty [GY95]

$$l_g(w) = \sum_{j=1}^p \frac{|w_j|}{|w_j| + b}$$

Capped L1 (or clipped L1, [Fan97, Zha09, ZSLS14])

$$ca_b(w) = \sum_{i=1}^n \min \left(1, \frac{|w_i|}{b} \right)$$

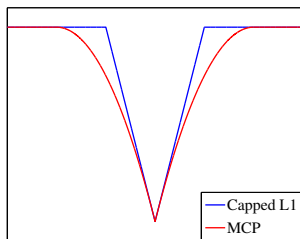
$$\lim_{b \rightarrow 0} ca_b(w) = c(w)$$

more approximations

MCP The minimum concave penalty [Zha10] (or firm shrinkage [GB97]), smoothed version of the capped L1, for $\gamma > 1$

$$MCP_{\lambda,\gamma}(t) = \lambda \int_0^t \left(1 - \frac{x}{\gamma\lambda}\right)_+ dx = \begin{cases} \lambda|t| - \frac{t^2}{2\gamma} & \text{if } |t| \leq \gamma\lambda \\ \frac{\gamma\lambda^2}{2} & \text{else.} \end{cases}$$

$$\lim_{\lambda\gamma \rightarrow 0, \frac{\gamma\lambda^2}{2} \rightarrow 1} MCP_{\lambda,\gamma}(w) = c(w)$$



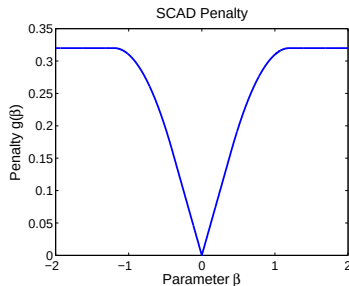
even more approximations

SCAD [FL01] smoothly clipped absolute deviation (SCAD) for $\gamma > 2$

$$SCAD_{\lambda, \gamma}(t) = \begin{cases} \lambda|t| & \text{if } t \leq \lambda \\ \frac{\gamma\lambda|t| - 0.5(t^2 + \lambda^2)}{\gamma - 1} & \text{if } \lambda < |t| \leq \gamma\lambda \\ \frac{\lambda^2(\gamma^2 - 1)}{2(\gamma - 1)} & \text{else.} \end{cases}$$

Based on a Bayesian argument, [Fan and Li, 2001] recommended to use the value of $\gamma = 3.7$.

In some sense, SCAD is a particular case of the MCP and thus can be seen as converging towards the counting function.



Non convex approximation of the counting function c

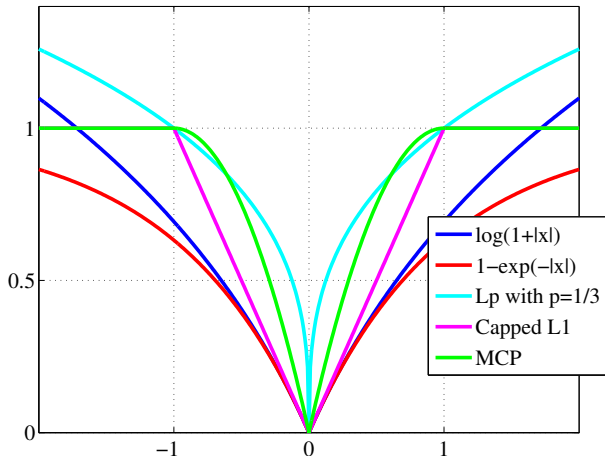


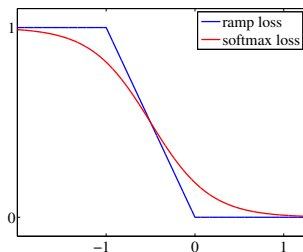
Figure : illustration of some functions non convex and non differentiable at $x = 0$.

Continuous approximation of the 0/1 loss function ℓ

The same kind of approximation can be made of the 0/1 loss function ℓ

The Ramp Loss function [CSWB06]: non-convexity can provide scalability advantages over convexity

The soft max loss (its continuous counterpart)



How to analyze these Continuous approximation

A good penalty function [AGN11]

- avoids excessive bias (unbiasedness),
- that forces sparse solutions to reduce model complexity (sparsity) and,
- that is continuous in the data to avoid unnecessary variation (stability).
- Moreover, from the computational viewpoint, penalty functions should be chosen in a way that the resulting optimization problem is easily solvable.

The shrinkage point of view

Approximation and shrinkage: 1d case

The particular case of the one dimensional penalized regression

$$\min_{w \in \mathbb{R}} \frac{1}{2} \|w\mathbf{x} - \mathbf{y}\|^2 + \lambda \mathbf{pen}(|w|)$$

$$\hat{w} = \frac{\mathbf{x}^t \mathbf{y}}{\mathbf{x}^t \mathbf{x}} = \mathbf{x}^t \mathbf{y} \text{ assuming } \mathbf{x}^t \mathbf{x} = 1$$

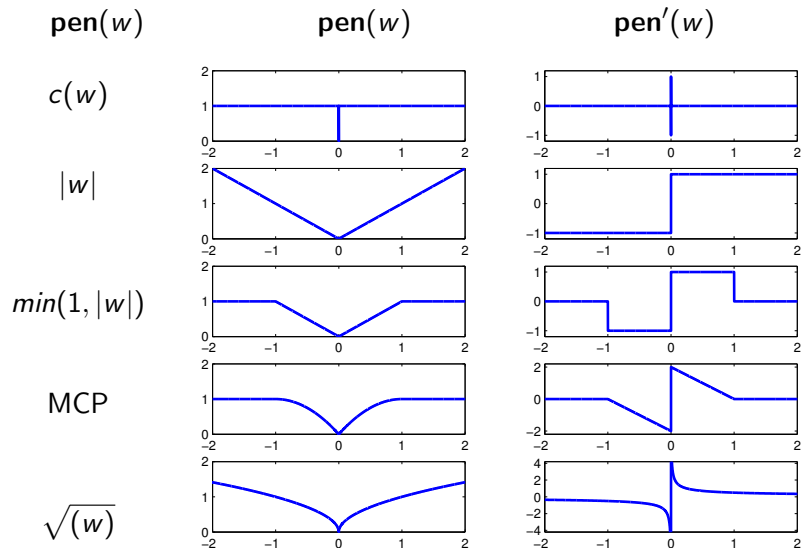
At the optimum, if $\mathbf{pen}'(|w|)$ exists, $\hat{w}_p$ the solution of the one dimensional penalized regression

$$\hat{w}_p \mathbf{x}^t \mathbf{x} - \mathbf{x}^t \mathbf{y} + \lambda \mathbf{pen}'(|w_p|) = 0 \qquad \hat{w}_p = \underbrace{\left(1 - \lambda \frac{\mathbf{pen}'(|w|)}{\hat{w}}\right)}_{\mathbf{shr}_\lambda(\hat{w})} \hat{w}$$

The solution is unique, The differentiability will be studied next,

Note that if $\mathbf{x} = 1$ then $\hat{w} = \mathbf{y}$

Example of penalties and associated differentials



Example of penalties and associated shrinkage functions

	$\text{pen}(t)$	$\text{pen}'(t)$	$\text{shr}_\lambda(\hat{w})$
Hard shr	$c(t)$	$0, t \neq 0$	$\begin{cases} 0 & \text{if } \hat{w} < \lambda \\ \hat{w} & \text{else} \end{cases}$
Soft shr	$ t $	$\text{sgn}(t), t \neq 0$	$\begin{cases} 0 & \text{if } \hat{w} < \lambda \\ \hat{w} - \text{sgn}(\hat{w})\lambda & \text{else} \end{cases}$
Capped L1	$\min(1, t)$	$\text{sgn}(t), t \leq \lambda$	$\begin{cases} 0 & \text{if } \hat{w} < \lambda \\ \hat{w} - \text{sgn}(\hat{w})\lambda & \text{if } \lambda < \hat{w} < \gamma\lambda \\ \hat{w} & \text{else} \end{cases}$
Firm shr	MCP		$\begin{cases} 0 & \text{if } \hat{w} < \lambda \\ \frac{\gamma(\hat{w} - \text{sgn}(\hat{w})\lambda)}{\gamma - \text{sgn}(\hat{w})} & \text{if } \lambda < \hat{w} < \gamma\lambda \\ \hat{w} & \text{else} \end{cases}$
SCAD			$\begin{cases} 0 & \text{if } \hat{w} < \lambda \\ \hat{w} - \text{sgn}(\hat{w})\lambda & \text{if } \lambda < \hat{w} < 2\lambda \\ \frac{(\gamma-2)\hat{w} - \text{sgn}(\hat{w})\gamma\lambda}{\gamma - 2\text{sgn}(\hat{w})} & \text{if } 2\lambda < \hat{w} < \gamma\lambda \\ \hat{w} & \text{else} \end{cases}$
nn-garrote			$\begin{cases} 0 & \text{if } \hat{w} < \lambda \\ \hat{w} - \text{sgn}(\hat{w})\frac{\lambda}{\hat{w}} & \text{else} \end{cases}$

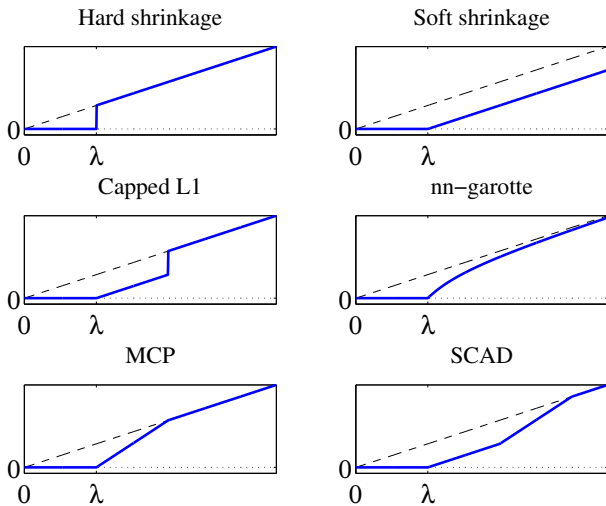


Figure : Example of shrinking functions.

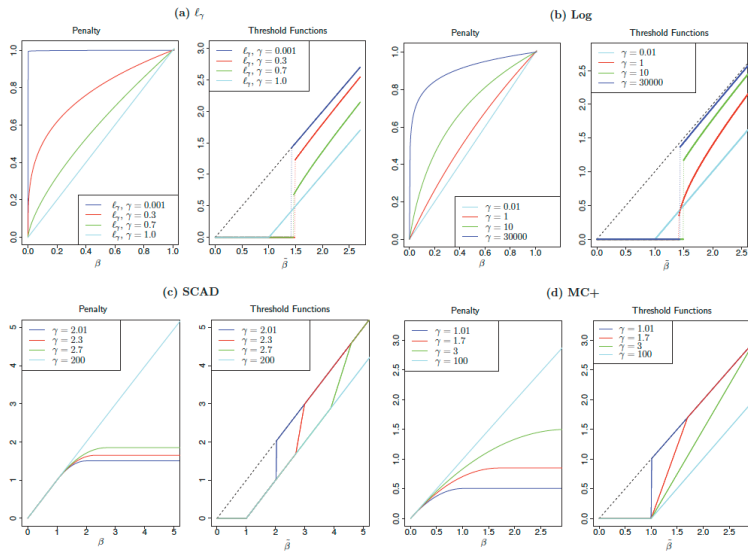


Figure : Example of shrinking functions with their corresponding penalties (From Sparset []).

The optimal shrinkage

Some estimator are directly defined based their shrinkage property such as the positive-part James–Stein estimator

$$\hat{w}_{js} = \max\left(0, \left(1 - \frac{(p-2)\sigma^2}{\|w\|^2}\right)w\right)$$

The non negative garrote:

$$\hat{w}_{nn} = \max\left(0, \left(1 - \frac{\lambda}{\hat{w}^2}\right)\hat{w}\right)$$

with the same shrinking behavior as the one of the nn-garrote 3

The best shrinkage

According to [FL01], a good shrinkage, just like a good penalty, should

- bring some sparsity,
- be unbiased and
- should be continuous.
- It should also be associated with a scaling procedure.

For instance the L_q bridge shrinking function is not continuous when $q < 1$.

As it is, shrinkage is simple, explicative, but it does not scale because it depend on $\hat{w}$.

Non convex approximation of the ranking function r

Assume $Y = U M V^t$ is the SDV decomposition. A non convex approximation of the solution of the penalized rank minimization can be given through any shrinking function by

$$\hat{X} = U \text{shr}_\lambda(M) V^t = \sum_{i=1}^{\min(n,p)} \text{shr}_\lambda(\mu_i) U_{i\bullet} V_{i\bullet}^t.$$

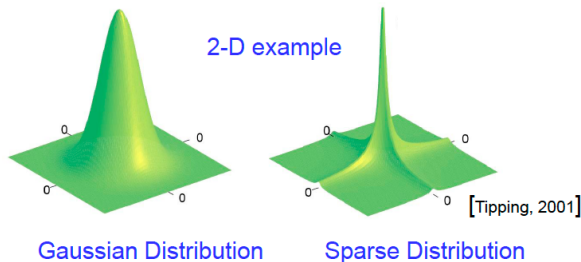
with for instance:

$\text{shr}_\lambda(\mu_i) =$ hard, soft, adaptive soft, MCP, SCAD, nn-garotte...

Can be efficient if there is an efficient way to compute the k largest singular values.

Non-Convex Bayesian Alternatives

- use A prior distribution $\Pr(w)$ that favors sparse coefficients (cf Tipping, Laplacian distribution) connection with “sparse Bayesian” modeling as shown by Tipping (2001).
- use Variational Bayes: Compute a posterior approximation that allows us to perform tractable inference
- Alternative non-convex MAP algorithms are possible,



Continuous relaxation

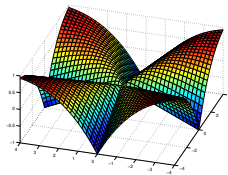
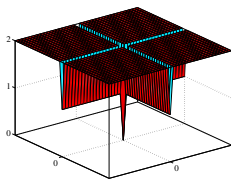
Non-convex sparsity-inducing penalties have recently received considerable attentions in sparse learning. Recent theoretical investigations have demonstrated their superiority over the convex counterparts in several sparse learning settings. [GZL⁺13].

Non-convex non-differentiable optimization remains a big challenge

- non convexity may provide multiple local minima
 - ▶ Need for global optimisation
- non differentiability
- computational efficiency: computationally tractable approximate methods are needed

Road map

- 1 Parcimonious and non-convex learning problems
- 2 Mixed integer programming for machine learning
- 3 Continuous relaxations
- 4 Nonsmooth and nonconvex optimization tools**
- 5 Optimization algorithm for nonconvex & nondifferential functionals
- 6 A case study: linear regression with MCP



Unconstrained minimization problem

$$\begin{aligned} f : \mathbb{R}^p &\longrightarrow \mathbb{R} \\ w &\longmapsto f(w) \end{aligned}$$

Unconstrained minimization

$$\min_{w \in \mathbb{R}^p} f(w)$$

Example (MCP regularization)

$$\min_{w \in \mathbb{R}^p} \underbrace{\frac{1}{2} \|Xw - y\|^2 + \lambda \sum_{j=1}^p MCP_{\lambda, \gamma}(|w_j|)}_{f(w)}$$

The program:

$$\min_{w \in \mathbb{R}^p} f(w)$$

The program:

- existence of optimal solutions,
- characterization of optimal solutions,
- algorithms for computing optimal solutions: how to find a descent direction (if it exists)

The concept of derivative is fundamental

Existence of a solution

Definition (local minima)

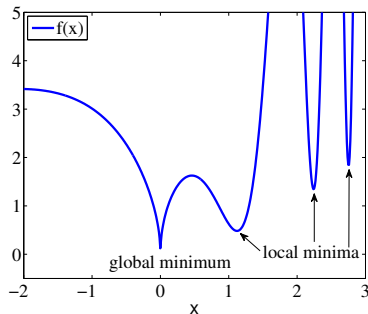
w^* is a local minima for f if there exists $\varepsilon > 0$ such that

$$f(w^*) \leq f(w) \quad \forall w \text{ with } \|w - w^*\| \leq \varepsilon$$

Definition (global minimum)

w^* is a global minimum for f if

$$f(w^*) \leq f(w) \quad \forall w \in \mathbb{R}^p$$



The question of the existence of a global minima to this unconstrained minimization problem require some definition.

Definition (l.s.c.)

a real valued function f is lower semi-continuous (l.s.c.) at some point w if for every sequence $\{w_k, k \in \mathbb{N}\}$ that converges to w

$$f(w) \leq \liminf_{k \rightarrow \infty} f(w_k).$$

The 0/1 loss function, the counting function and the rank function are l.s.c.

Definition (coercive)

a real valued function f is coercive if for every sequence $\{w_k, k \in \mathbb{N}\}$ such that $\lim_{k \rightarrow \infty} \|w_k\| = \infty$

$$\lim_{k \rightarrow \infty} f(w_k) = \infty.$$

Proposition

Weierstrass' theorem: existence of a solution (sufficient condition). If f is l.s.c. and coercive or admits a non empty bounded level set, then there exists a global minimizer for f .

Gradient

The gradient is a generalization of the usual concept of derivative of a function in one dimension to a function in several dimensions.

$$\begin{array}{ccc} f : & \mathbb{R}^p & \longrightarrow \mathbb{R} \\ & \mathbf{w} & \longmapsto f(\mathbf{w}) \end{array}$$

Assume f is differentiable in the sense that all its partial derivatives $\frac{\partial f}{\partial w_i}$ exists.

Definition (gradient)

The gradient $\nabla f(\mathbf{w})$ of a function f at point $\mathbf{w}$ is the vector whose components are the partial derivatives of f

Example

The least square

$$f_1(\mathbf{w}) = \|A\mathbf{w} - \mathbf{y}\|^2$$

$$\nabla f_1(\mathbf{x}) = 2A^t(A\mathbf{w} - \mathbf{y})$$

Gradient's properties

Proposition

If f is convex and differentiable, then

$$f(\mathbf{w} + h) \geq f(\mathbf{w}) + \nabla f(\mathbf{w})^t h, \quad \forall h \in \mathbb{R}^p$$

consequence global solution

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} f(\mathbf{w}) \iff \nabla f(\mathbf{w}^*) = 0$$

On the other way round, if f is such that

$$\nabla f(\mathbf{w})^t h \Rightarrow f(\mathbf{w} + h) \geq f(\mathbf{w}), \quad \forall h \in \mathbb{R}^p$$

The gradient provides:

- optimality condition (local or global),
- a descent direction,
- handy computation rules: sum and chain rule.

What if f is not differentiable? $f_2(x) = |x|$

Definition (subgradient)

A vector $d \in \mathbb{R}^p$ is a subgradient of a function f at point w if

$$f(w + h) \geq f(w) + d^t h, \quad \forall h \in \mathbb{R}^p$$

Definition (subdifferential)

The subdifferential $\partial f(w)$ of a function f at point w is the set (possibly empty) of all its subgradients

$$\partial f(w) = \{g \in \mathbb{R}^p \mid f(w + d) \geq f(w) + g^t d, \quad \forall d \in \mathbb{R}^p\}$$

Example

$$n = 1$$

$$f_2(x) = |x|$$

$$f_3(x) = \max(0, 1 - x)$$

$$\partial f_2(0) = \{g \in \mathbb{R} \mid -1 \leq g \leq 1\}$$

$$\partial f_3(1) = \{g \in \mathbb{R} \mid -1 \leq g \leq 0\}$$

Convexity of f implies that it has at least one supporting hyperplane at every point of $\mathbb{R}^p$.

Proposition

If f is convex, then $\partial f(\mathbf{w}) \neq \emptyset$. Furthermore, If f is differentiable, $\nabla f(\mathbf{w})$ is a subgradient of f at $\mathbf{w}$ that is

$$\partial f(\mathbf{w}) = \{\nabla f(\mathbf{w})\}.$$

Finally, it characterizes the global solution

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} f(\mathbf{w}) \iff 0 \in \partial f(\mathbf{w}^*).$$

What if f is not convex?

Example

$$f_4(x) = \log(1 + |x|) \quad \partial f_4(0) = 0$$

Dini Hadamar subgradient

A vector $g \in \mathbb{R}^p$ is a (global) subgradient of f at w if

$$f(w + d) \geq f(w) + g^t d, \quad \forall d \in \mathbb{R}^p$$

A necessary condition, for w^* being a global minima is that

$$g^t d \geq 0 \quad \forall d \in \mathbb{R}^p$$

A necessary condition, for w^* being a local minim is that

$$\frac{f(w^* + td) - f(w^*)}{t} \geq g^t d + o(t) \geq 0, \quad \forall d \text{ with } \|d\| = 1 \text{ and } t \leq \varepsilon$$

Definition (Dini Hadamar subgradient)

A vector $g \in \mathbb{R}^p$ is a Dini Hadamar subgradient of f at w , if

$$\lim_{t \rightarrow 0} \frac{f(w^* + td) - f(w^*)}{t} \geq g^t d, \quad \forall d \in \mathbb{R}^p$$

Example of Dini Hadamar subgradient

Example

$$f(w) = \begin{cases} -w & \text{if } w < 0 \\ \sqrt{w} & \text{else} \end{cases}$$

the subgradients at $w = 0$ are $-1 \leq g \leq 0$ and all $-1 \leq g$ are Dini Hadamar subgradients.

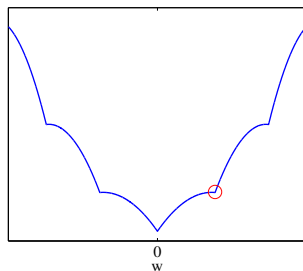
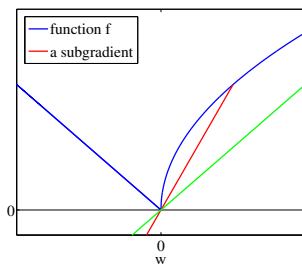


Figure : Illustration of the sugradient.

Directional derivative, Dini subgradient and subdifferential

Definition (directional derivative)

of a function f in the direction d at point w (if it exists)

$$Df(w, d) = \lim_{\varepsilon \rightarrow 0^+} \frac{f(w + \varepsilon d) - f(w)}{\varepsilon}$$

Definition (Dini-Hadamard subgradient)

A vector $g \in \mathbb{R}^p$ is a Dini-Hadamard subgradient of f at w if

$$g^\top d \leq Df(w, d), \quad \forall d \in \mathbb{R}^p.$$

Definition (Dini-Hadamard subdifferential)

of a function f at point w is the set of all its Dini-Hadamard subgradients

$$\partial_{DH} f(w) = \{g \in \mathbb{R}^p \mid Df(w, d) \geq d^\top g, \quad \forall d \in \mathbb{R}^p\}.$$

Example (from [BF08])

$$f_4(x) = \log(1 + |x|)$$

$$Df_4(0, d) = |d|$$

$$\partial_{DH} f_4(0) = [-1, 1]$$

$$f_5(x_1, x_2) = |x_1| - |x_2|$$

$$Df_5(0, d) = |d_1| - |d_2|$$

$$\partial_{DH} f_5(0) = \emptyset$$

$$f_6(x_1, x_2) = |x_1| + |x_2| - 1.25\sqrt{x_1^2 + x_2^2}$$

$$\partial_{DH} f_7(0) = \emptyset$$

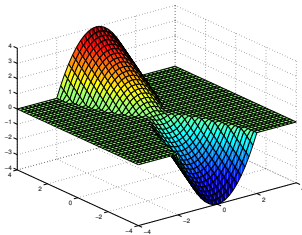
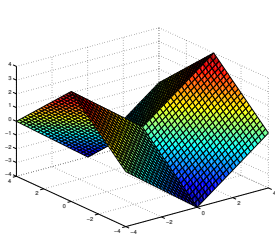


Figure : functions f_5 , f_6 and f_7

What if the Dini-Hadamard subdifferential is empty?

The Clarke subdifferential is a generalization of the subdifferential to non convex functions. It is initially defined for locally Lipschitz functions f as the convex hull of some generalized gradient and more precisely

Definition (locally Lipschitz continuity)

A function f is locally Lipschitz continuous, if for each w_0 there exist neighborhood V of w_0 and a constant $M_{w_0} \geq 0$ such that

$$|f(y) - f(w)| \leq M_{w_0} \|y - w\| \quad \forall (w, y) \in V^2$$

The three functions are not continuous so they are not locally Lipschitz continuous either. But some of their non convex relaxations are.

Example

$f_5(x_1, x_2) = |x_1| - |x_2|$ is locally Lipschitz continuous with $M = 1$. $\sqrt{x}$ and $\log(x)$ are not locally Lipschitz continuous at $w_0 = 0$, but $\log(x + \varepsilon)$ is.

Clarke subdifferential

Let f be a locally Lipschitz continuous function at w

Definition (Clarke directional derivative)

The Clarke directional derivative of f at point w in direction d is

$$D_c f(w, d) = \limsup_{y \rightarrow w, \varepsilon \rightarrow 0+} \frac{f(y + \varepsilon d) - f(y)}{\varepsilon}.$$

Definition (Clarke subgradient)

A vector $g \in \mathbb{R}^p$ is a Clarke subgradient of f at w if

$$g^\top d \leq D_c f(w, d), \quad \forall \quad d \in V.$$

Definition (Clarke subdifferential)

The Clarke subdifferential is the set of all its Clarke subgradient,

$$\partial_c f(w) = \{g \in \mathbb{R}^p \mid g^\top d \leq D_c f(w, d), \quad \forall d \in \mathbb{R}^p\}.$$

Clarke subdifferential's properties

Clarke's subdifferential can be seen as the closure of the set of limits of gradients.

Note that if f is convex, its Clarke's subdifferential coincide with its subdifferential and if f is differentiable coincide with its gradient

$$\partial_c f(w) = \{\nabla f(w)\}.$$

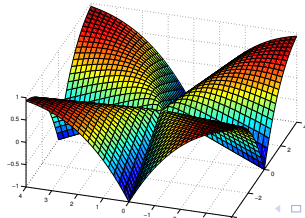
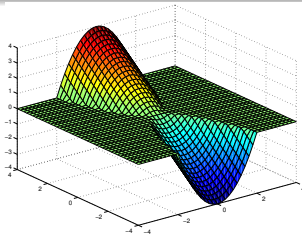
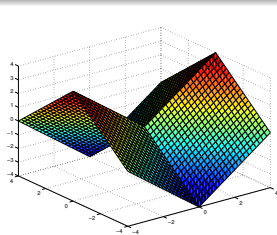
$$\partial_{DH} f(w) \subset \partial_c f(w)$$

nice calculus rules

Example of Clarke subdifferentials

Example (from [BF08])

$$\begin{array}{lll} f_5(x_1, x_2) = |x_1| - |x_2| & D_c f_5(0, d) = |d_1| + |d_2| & \partial_c f_5(0) = [-1, 1]^2 \\ f_6(x_1, x_2) & & \partial_c f_6(0) = \{0\} \times [0, 1] \\ f_7(x_1, x_2) & & \partial_c f_7(0) = \end{array}$$



More on subdifferentials

- the Lipschitz hypothesis can be removed [RW09]
- that there exist many different nonequivalent definition of subdifferentials [Pen96].
- Bolte definition: the subdifferential of f at w is the set of all directions d such that there exists two sequences w_k and d_k converging towards w and d and

$$f(y) \geq f(w_k) + d_k^t(u - w_k) + o(\|u - w_k\|) \quad \forall u \in \mathbb{R}^p$$

- The subdifferential of the counting function [Le13, theorem 2.9]

$$\partial_c c(x) = \{y \in \mathbb{R}^p \mid y_i = 0 \text{ when } x_i \neq 0\}$$

The subdifferential of the rank function [Le13, theorem 2.14]

Unconstrained optimization (Fermat's rule)

For f convex & differentiable, global minima

$$w^* = \arg \min_w f(w) \iff \nabla f(w^*) = 0$$

For f convex & nondifferentiable, global minima

$$w^* = \arg \min_w f(w) \iff 0 \in \partial f(w^*)$$

For f non convex & nondifferentiable, local minima, necessary condition

$$w^* = \arg \min_{w \in V(w^*)} f(w) \implies 0 \in \partial_c f(w^*)$$

or equivalently (sometimes referred as Oresme's rule)

$$w^* = \arg \min_{w \in V(w^*)} f(w) \implies D_c f(w^*, w) \geq 0, \quad \forall w \in V(w^*)$$

Fermat's rule and critical points

Table : Fermat's rule and critical (or stationary) points

	differentiable	non differentiable
cvx	$w^* = \arg \min_w f(w) \Leftrightarrow \nabla f(w^*) = 0$	$w^* = \arg \min_w f(w) \Leftrightarrow 0 \in \partial f(w^*)$
non cvx	$w^* = \arg \min_{w \in V(w_0)} f(w) \Rightarrow \nabla f(w^*) = 0$	$w^* = \arg \min_{w \in V(w_0)} f(w) \Rightarrow 0 \in \partial_c f(w^*)$

Definition (Clarke critical (or stationary) point)

Let f be a locally Lipschitz continuous function at w . A point w^* is called a (Clarke) critical point if

$$0 \in \partial_c f(w)$$

Local minima are critical points but the converse is not always true.

Local and global minima

Regularity:

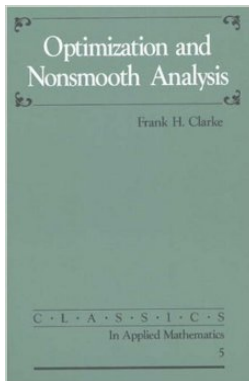
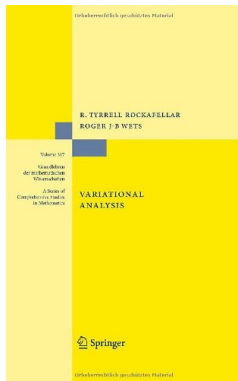
- A function f is regular in the sense of Clarke at $w \in \mathbb{R}^p$ if, $\forall d \in \mathbb{R}^p$, $Df(w, d) = D_c f(w, d)$.
- A function is convex like (invex) is a critical point is also a local minima.

Example (Minimizing the rank of a matrix (Hiriart-Urruty, 2013))

$$\begin{cases} \min_A & \text{rank}(A) \\ \text{s.t.} & A \in C \end{cases}$$

Every admissible point is a local minima [HU13]. Only global minimization matters.

Subdifferential calculus



Subdifferential general rules

Theorem (Multiplication and sum rule)

Let f be proper, lower-semicontinuous and Lipschitz near w , then $\forall a \in \mathbb{R}$

$$\partial_c af(w) = a\partial_c f(w).$$

where $aA = \{ax \mid w \in A\}$. Let g and h be proper, lower-semicontinuous and Lipschitz near w . Then if $f = g + h$

$$\partial_c f(w) \subset \partial_c g(x) \oplus \partial_c h(x)$$

where $A \oplus B = \{a + b \mid a \in A \text{ and } b \in B\}$.

In particular, for critical points

$$0 \in \partial_c f(w^*) \Rightarrow -\partial_c h(w^*) \cap \partial_c g(w^*) \neq \emptyset.$$

+ The Chain rule

Subdifferential calculus for DC functions

Theorem (DC sum rule [BF08])

Let $f = g - h$ with $g; h$ be proper, lower-semicontinuous and **convex**. Then

$$\partial_{DH}f(w) = \partial g(w) \ominus \partial h(w)$$

$\ominus$ being the Minkowski difference, $A \ominus B = \{x \in \mathbb{R}^p \mid \{x\} \oplus B \subset A\}$ [Mar00].

Theorem (DC optimality, proposition 5.1. in [BF08])

Let $g; h$ be proper, lower-semicontinuous and convex. Then if $f = g - h$ and w^* a local minima

$$0 \in \partial_{DH}f(w^*) \quad \implies \quad \exists(a, b) \in (\partial g(w), \partial h(w)) \text{ such that } a - b = 0$$

Note also that this condition is also **sufficient** for some classes of DC programs such as **DC polyhedral** programs and **locally convex** ones at the solution [TTA10].

DC functions are quasidifferentiable

A remark regarding operations between sets

- addition (Minkovski) $A \oplus B = \{a + b \in \mathbb{R}^p \mid a \in A; b \in B\}$
- multiplication $\lambda A = \{\lambda a \in \mathbb{R}^p \mid a \in A\}$
- subtraction
 - ▶ algebraic difference $A - B = \{a - b \in \mathbb{R}^p \mid a \in A; b \in B\}$
 - ▶ geometric difference $A \ominus B = \{x \in \mathbb{R}^p \mid \{x\} \oplus B \subset A\}$

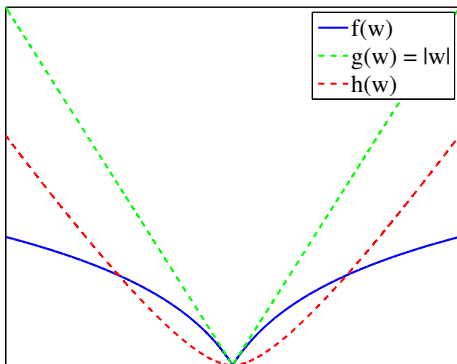
Example

A	B	$A - B$	$A \ominus B$	
$[1, 3]$	$\{2\}$	$[-1, 1]$	$[-1, 1]$	
$[0, 1]$	$[0, 1]$	$[-1, 1]$	$\{0\}$	$\leftarrow A \ominus A = \{0\}$
$[0, 1]$	$[-1, 0]$	$[-2, 0]$	$\{1\}$	
$[1, 2]$	$[-1, 1]$	$[-3, 0]$	$\emptyset$	

Example (the Log-sum approximation)

A function f is said to be locally convex on at w if its directional derivative $Df(w, d)$ exists and is convex in d . For instance $f_4(w) = \log(1 + |w|)$ is locally convex since $Df_4(w, d) = \frac{|d|}{1+|w|}$. It is also DC for $w > 0$ since

$$f_4(w) = \underbrace{|w|}_g - \underbrace{(|w| - \log(1 + |w|))}_h$$



Example (Capped L1)

$$f(w) = \begin{cases} |x| & \text{if } |x| \leq 1 \\ 1 & \text{else} \end{cases} = |x| - \underbrace{\begin{cases} 0 & \text{if } |x| \leq 1 \\ |x| & \text{else} \end{cases}}_{g-h}$$

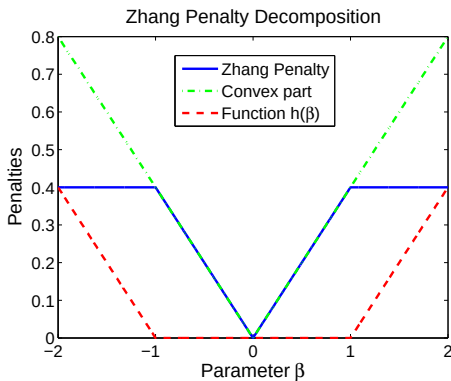
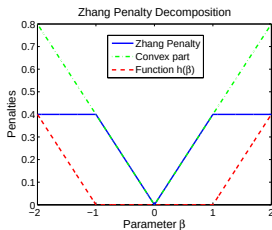


Figure : decomposition of the capped L1 penalty



$$\partial g(w) = \begin{cases} -1 & \text{if } w < 0 \\ [-1, 1] & \text{if } w = 0 \\ 1 & \text{if } w > 0 \end{cases} \quad \partial h(w) = \begin{cases} -1 & \text{if } w < -1 \\ [-1, 0] & \text{if } -1 < w < 0 \\ 0 & \text{if } |w| < 1 \\ [0, 1] & \text{if } 0 < w < 1 \\ 1 & \text{if } w > 1 \end{cases}$$

so that

$\partial_{DH} f(w)$	$\partial_c f(w)$	
0	0	if $w < -1$
$\emptyset$	$[-1, 0]$	if $w = -1$
-1	-1	if $-1 < w < 0$
$[-1, 1]$	$[-1, 1]$	if $w = 0$
1	1	if $0 < w < 1$
$\emptyset$	$[0, 1]$	if $w = 1$
0	0	if $w > 1$

Another useful particular case

Theorem (Sum rule with g smooth, theorem 10.1 in [RW09])

Let $f; h$ be proper, lower-semicontinuous, g is differentiable at w . Then if $f = g + h$

$$\partial_c f(w) = \{\nabla g(w)\} \oplus \partial_c h(w)$$

in particular, $0 \in \partial_c f(w^*)$ is equivalent to $-\nabla g(w^*) \in \partial_c h(w^*)$

Example (proximal mapping and shrinkage)

For a given $\lambda > 0$ and $x_0 \in \mathbb{R}^p$, let

$$f(w) = \frac{1}{2\lambda} \|w - w_0\| + h(w)$$

if w^* is a minimizer of f then

$$0 \in \partial_c f(w^*) \quad \Leftrightarrow \quad -\frac{1}{\lambda}(w^* - w_0) \in \partial_c h(w^*)$$

so that

$$w_0 \in (w^* + \lambda \partial_c h(w^*)) \quad \Leftrightarrow \quad w^* \in (I + \lambda \partial_c h)^{-1}(w_0)$$

Theorem (Toland, 1979, cited in [GZL⁺13])

Let g differentiable, h_1 and h_2 be proper, lower-semicontinuous and convex. Then if $f = g + h_1 - h_2$

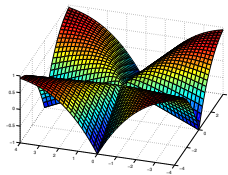
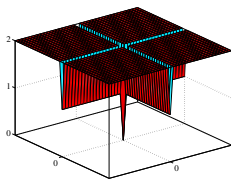
$$0 \in \partial_c f(w^*) \quad \Rightarrow \quad -\nabla g(w^*) \in \partial h_1(w^*) \ominus \partial h_2(w^*)$$

Useful optimization tools

- general case: chasing critical points.
- for specific non convex and non differential function critical points are local minima
 - ▶ DC polyhedral programs
 - ▶ DC locally convex
 - ▶ differentiable + convex

Road map

- 1 Parcimonious and non-convex learning problems
- 2 Mixed integer programming for machine learning
- 3 Continuous relaxations
- 4 Nonsmooth and nonconvex optimization tools
- 5 Optimization algorithm for nonconvex & nondifferential functionals
- 6 A case study: linear regression with MCP



Nonconvex nonsmooth continuous optimization algorithms

- sub gradient
stochastic sub gradient: Yann tomorrow (when differentiable)
- homotopy methods: the regularization path algorithms and continuation methods
- Proximal
GIST
 - ▶ accelerations
 - ▶ proximal newton
- Component wise optimization and block coordinate descent
SparseNet
- DCA, CCCP, LLA (QLA)
- Majorization Minimization
- Active set

Subgradient descent

Algorithm 1 Subgradient algorithm

Data: w initialization

Result: w

initialize: find a feasible w

while *not converged* **do**

$d \leftarrow$ a subgradient: $d \in \partial_c f(w)$

$\rho \leftarrow$ choose a stepsize

$w \leftarrow w - \rho d$

 check that the cost decrease

end

Opposite direction of subgradient is NOT always a descent direction.
However it may converge toward a stationary point (cf Bretsekas).

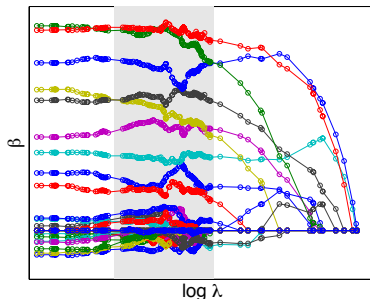
Greedy and homotopy methods: regularization path

Use a greedy algorithm to replace the exhaustive search of $w(\lambda)$, $\lambda \in \mathbb{R}^+$

- Greedy algorithms (forward/backward)
- Continuation: prediction correction

$$w(\lambda^{k+1}) = \varphi(w(\lambda^k), \lambda^k)$$

- Regularization path (SCAD and MCP) [Breheny and Huang, 2011]
- Parametric programming



Proximals approaches

- invented for convex non differentiable functions
- modified gradient
- adapted for certain non convex
- mini road map
 - ▶ Proximity operator
 - ▶ Proximal gradient algorithm
 - ▶ some properties

adapted from Boyd & Vandenbergue lecture 18 on Proximal gradient method

Proximity operator

Definition (Proximity operator)

The Proximity operator of a function h is:

$$\begin{aligned} \mathbf{prox}_h : \mathbb{R}^p &\longrightarrow \mathbb{R} \\ w &\longmapsto \mathbf{prox}_h(w) = \arg \min_{u \in \mathbb{R}^p} h(u) + \frac{1}{2} \|u - w\|^2 \end{aligned}$$

Example

$h(w) = 0$	$\mathbf{prox}_h(w) = w$	
$h(w) = \mathbb{I}_C(w)$	$\mathbf{prox}_h(w) = \arg \min_{u \in C} \frac{1}{2} \ u - w\ ^2$	projection
$h(w) = \rho \mathbf{pen}_\lambda(w)$	$\mathbf{prox}_h(w) = \mathbf{shr}_{\rho\lambda}(w)$	shrinkage

Proximal gradient algorithm

Assume the splitting $\min_{\mathbf{w} \in \mathbb{R}^p} f(\mathbf{w}) = g(\mathbf{w}) + h(\mathbf{w})$

- $g(\mathbf{w})$ differentiable, gradient Lipschitz
- $h(\mathbf{w})$ such that the **prox** exists

The Proximal gradient sequence

$$\mathbf{w}^{k+1} = \mathbf{prox}_{\rho h}(\mathbf{w}^k - \rho \nabla g(\mathbf{w}^k))$$

Algorithm 2 Proximal gradient algorithm

Data: $\mathbf{w}$ initialization

Result: $\mathbf{w}$

while *not converged* **do**

$\mathbf{g} \leftarrow \nabla g(\mathbf{w})$

$\rho \leftarrow$ choose a stepsize

$\mathbf{d} \leftarrow \mathbf{w} - \rho \mathbf{g}$

$\mathbf{w} \leftarrow \mathbf{prox}_{\rho h}(\mathbf{d})$

end

Examples

The Proximal gradient sequence

$$\mathbf{w}^{k+1} = \mathbf{prox}_{\rho h}(\mathbf{w}^k - \rho \nabla g(\mathbf{w}^k))$$

Example

$$h(\mathbf{w}) = 0$$

$$\mathbf{prox}_h(\mathbf{w}) = \mathbf{w}$$

The gradient method

$$h(\mathbf{w}) = \mathbb{I}_C(\mathbf{w})$$

$$\mathbf{prox}_h(\mathbf{w}) = \text{projection}$$

The projected gradient

$$h(\mathbf{w}) = \rho \mathbf{pen}_{\lambda}(\mathbf{w})$$

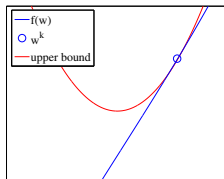
$$\mathbf{prox}_h(\mathbf{w}) = \mathbf{shr}_{\rho\lambda}(\mathbf{w})$$

Iterative shrinkage

Interpretation of the algorithm

Assume the splitting $\min_{\mathbf{w} \in \mathbb{R}^p} f(\mathbf{w}) = g(\mathbf{w}) + h(\mathbf{w})$

$$\begin{aligned} \mathbf{w}^{k+1} &= \mathbf{prox}_{\rho h}(\mathbf{w}^k - \rho \nabla g(\mathbf{w}^k)) \\ &= \arg \min_{\mathbf{u} \in \mathbb{R}^p} \rho h(\mathbf{u}) + \frac{1}{2} \|\mathbf{u} - \mathbf{w}^k + \rho \nabla g(\mathbf{w}^k)\|^2 \\ &= \arg \min_{\mathbf{u} \in \mathbb{R}^p} h(\mathbf{u}) + \frac{1}{2\rho} \|\mathbf{u} - \mathbf{w}^k + \rho \nabla g(\mathbf{w}^k)\|^2 \\ &= \arg \min_{\mathbf{u} \in \mathbb{R}^p} h(\mathbf{u}) + (\mathbf{u} - \mathbf{w}^k)^t \nabla g(\mathbf{w}^k) + \frac{1}{2\rho} \|\mathbf{u} - \mathbf{w}^k\|^2 \\ &= \arg \min_{\mathbf{u} \in \mathbb{R}^p} h(\mathbf{u}) + \underbrace{g(\mathbf{w}^k) + (\mathbf{u} - \mathbf{w}^k)^t \nabla g(\mathbf{w}^k) + \frac{1}{2\rho} \|\mathbf{u} - \mathbf{w}^k\|^2}_{\text{a quadratic local model of } g \text{ near } \mathbf{w}^k} \end{aligned}$$



Theorem (quadratic upper bound)

if g is gradient Lipschitz with a constant $L = \frac{1}{\rho}$, then

$$\forall u \in \mathbb{R}^p \quad f(u) \leq h(u) + g(w^k) + (u - w^k)^t \nabla g(w^k) + \frac{1}{2\rho} \|u - w^k\|^2$$

Proof.

$$g(u) - g(w^k) = \int_0^1 (u - w^k)^t \nabla g(w^k + t(u - w^k)) dt$$

with $d = (u - w^k)$

$$\begin{aligned} g(u) &= g(w^k) + d^t \nabla g(w^k) + \int_0^1 d^t (\nabla g(w^k + td) - \nabla g(w^k)) dt \\ &\leq g(w^k) + d^t \nabla g(w^k) + \int_0^1 \|d\| \|\nabla g(w^k + td) - \nabla g(w^k)\| dt \\ &\leq g(w^k) + d^t \nabla g(w^k) + \int_0^1 tL \|d\|^2 dt \\ &\leq g(w^k) + (u - w^k)^t \nabla g(w^k) + \frac{L}{2} \|u - w^k\|^2 \end{aligned}$$

Decreasing sequence

Theorem

For a proper choice of the step size, the Proximal gradient sequence is decreasing

$$w^{k+1} = \mathbf{prox}_{\rho h}(w^k - \rho \nabla g(w^k))$$

We have by definition of the prox

$$h(w^{k+1}) + (w^{k+1} - w^k)^t \nabla g(w^k) + \frac{1}{2\rho} \|w^{k+1} - w^k\|^2 \leq h(w^k)$$

and since g is gradient Lipchitz

$$g(w^{k+1}) \leq g(w^k) + (w^{k+1} - w^k)^t \nabla g(w^k) + \frac{L}{2} \|w^{k+1} - w^k\|^2$$

so that since $f(w^{k+1}) = h(w^{k+1}) + g(w^{k+1})$

$$f(w^{k+1}) \leq f(w^k) - \frac{\rho - L}{2} \|w^{k+1} - w^k\|^2$$

Fixed point and critical point

The Proximal gradient sequence

$$\begin{aligned} w^* &= \mathbf{prox}_{\rho h}(w^* - \rho \nabla g(w^*)) \\ &= \arg \min_{u \in \mathbb{R}^p} h(u) + (u - w^*)^t \nabla g(w^*) + \frac{1}{2\rho} \|u - w^*\|^2 \end{aligned}$$

so that

$$0 \in \partial_c h(w^*) \oplus \nabla g(w^*) = \partial_c f(w^*)$$

and thus a fixed point of the proximal sequence is a critical point of f

Prox summary

- General iterative shrinkage and thresholding procedure (GIST)
- convergence proofs
- improving the convergence: FISTA and many others
- improvement: Newton proximal
- more improvement with randomization
- Matlab toolbox (GIST)
- general framework: more applications

Component wise optimization

Coordinate wise optimization A Gauss-Seidel approach (like SMO)

1d problems are easier to solve

Algorithm 3 Coordinate wise optimization

Data: w, λ, γ initialization

Result: w

while *not converged* **do**

$\text{ind} \leftarrow \text{randperm}(p)$

for $j = 1, p$ **do**

$w_{\text{ind}(j)} \leftarrow \text{Shrink}(w_{\text{ind}(j)}, \lambda, \gamma)$

end

end

Available software: SparseNet

<http://redwood.berkeley.edu/bruno/sparsenet/>

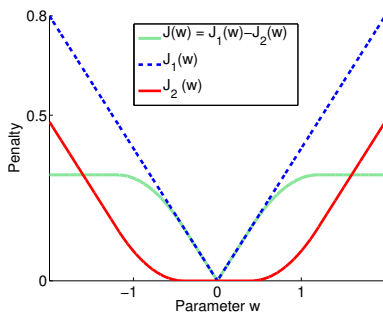
Under some conditions limit point(s) of the sequence of coordinate-wise updates are local minimizers (critical points).

Alternating Minimization

Difference of Convex functions

DC function

- Let $J_1(w), J_2(w) : \mathcal{C} \rightarrow]-\infty, +\infty]$ two **convex, proper and lower semi-continuous functions**
- $J(w)$ is a DC function if it can be expressed as $J(w) = J_1(w) - J_2(w)$.



Properties of DC functions

Non-uniqueness of a DC decomposition

- Let $J(w) = J_1(w) - J_2(w)$ a DC function
- Let $g(w)$ a convex, proper and lsc function
- J can be expressed as $J(w) = (J_1(w) + g(w)) - (J_2(w) + g(w))$

Linear combination

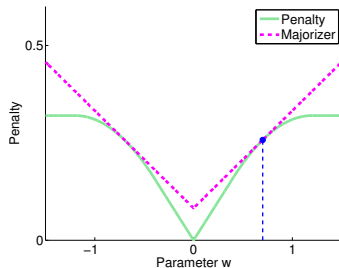
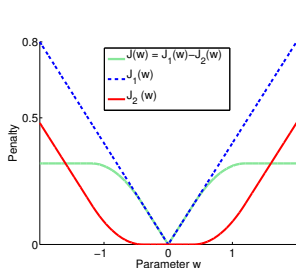
- Let $J_k(w) = J_{k,1}(w) - J_{k,2}(w)$, $k = 1, \dots, M$ being DC functions
- Any function $\sum_{k=1}^M \beta_k J_k(w)$ with $\beta_k \in \mathbb{R}$ is a DC function

Properties of DC functions

Convex majorization

- Let $\partial J_2(w_t) = \{\alpha_t \in \mathbb{R}^d, J_2(w) \geq J_2(w_t) + \langle w - w_t, \alpha_t \rangle, \forall w \in \mathbb{R}^d\}$ the subdifferential of J_2 at w_t .
- A convex majorization function of $J(w) = J_1(w) - J_2(w)$ at w_t is

$$J(w) \leq J_1(w) - J_2(w_t) - \langle w - w_t, \alpha_t \rangle$$



DC Algorithm

Principle: successive convex relaxations

- At each iteration t , define the convex majorization function

$$J_{cvx}(w) = J_1(w) - J_2(w_t) - \langle w - w_t, \alpha_t \rangle \quad \text{with} \quad \alpha_t \in \partial J_2(w_t)$$

- Next solution: $w_{t+1} = \arg \min_w J_{cvx}(w)$

Algorithm for solving $\min_w J_1(w) - J_2(w)$

Set $t = 0$, initialize $w_t \in \text{dom} J_1$

repeat

 Select $\alpha_t \in \partial J_2(w_t)$

 Define $J_{cvx}(w)$ and solve $w_{t+1} = \arg \min_w J_{cvx}(w)$

$t = t + 1$

until convergence

Convergence

Main Results

Assume $J(w) = J_1(w) - J_2(w)$ a coercive function with J_1, J_2 , lsc proper convex functions such as $\text{dom} J_1 \subseteq \text{dom} J_2$. It holds

- the sequence $\{w_t\}$ is well defined or equivalently $\text{dom } \partial J_1 \subseteq \text{dom } \partial J_2$
- the sequence $\{J(w_t)\}$ is monotonically decreasing
- if the minimum of J is finite, every limit point $\hat{w}$ of the bounded sequence $\{w_t\}$ (J being coercive) is a critical point of J and satisfies the local optimality condition

Convergence

Decrease of the objective function

- Establish that $J_1(w_{t+1}) - J_2(w_{t+1}) \leq J_1(w_t) - J_2(w_t)$ for all t
- $\alpha_t \in \partial J_2(w_t) = \{\alpha, J_2(w) \geq J_2(w_t) + \langle w - w_t, \alpha_t \rangle\}$ implies:
 $-J_2(w) \leq -J_2(w_t) + \langle w_t - w, \alpha_t \rangle \quad \forall w$, hence
 $-J_2(w_{t+1}) \leq -J_2(w_t) + \langle w_t - w_{t+1}, \alpha_t \rangle$
 $J_1(w_{t+1}) - J_2(w_{t+1}) \leq J_1(w_{t+1}) - J_2(w_t) + \langle w_t - w_{t+1}, \alpha_t \rangle \quad (i)$
- $w_{t+1} = \arg \min_w J_1(w) - J_2(w_t) - \langle w - w_t, \alpha_t \rangle$ leads to
 $J_1(w_{t+1}) - J_2(w_t) + \langle w_t - w_{t+1}, \alpha_t \rangle \leq J_1(w_t) - J_2(w_t) \quad (ii)$
- (i) and (ii) imply the desired result $J(w_{t+1}) \leq J(w_t)$

Links to other methods

- Convex-Concave procedure (CCCP) equivalent to DC procedure for differentiable functions J_1 and J_2
- DC Algorithm is a Majorization-Minimization procedure
- Multistage convex relaxation approach based on concave duality

Common feature

- Bound the objective function by a convex relaxation
- Reduce the bound by minimizing the relaxation function to yield a new solution

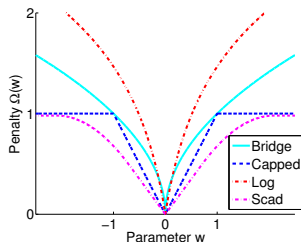
DC algorithm in play

Optimization problem

$$\min_{\mathbf{w} \in \mathbb{R}^p} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \sum_{j=1}^d \text{pen}(|w_j|)$$

Non-convex penalties

- Bridge: $\text{pen}(w_j) = |w_j|^p$, $p \in (0, 1)$
- Log: $\text{pen}(w_j) = \log(|w_j|^p + \epsilon)$,
- Capped ℓ_1 : $\text{pen}(w_j) = \min(\eta, |w_j|)$
- Geman
- nn-garotte
- MCP, SCAD



DC decomposition

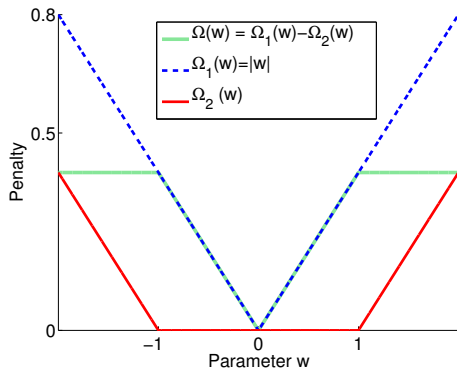
DC Decomposition of the penalty

- $\text{pen}(|w_j|) = \text{pen}_1(|w_j|) - \text{pen}_2(|w_j|)$
- $\text{pen}_1(|w_j|) = |w_j|$ and $\text{pen}_2(|w_j|) = |w_j| - \text{pen}(|w_j|)$

Example

For capped ℓ_1 penalty we have

- $\text{pen}(|w_j|) = \min(\eta, |w_j|)$
- $\text{pen}_2(|w_j|) = \max(0, |w_j| - \eta)$



DC decomposition

DC Decomposition of the penalty

- $\text{pen}(|w_j|) = \text{pen}_1(|w_j|) - \text{pen}_2(|w_j|)$
- $\text{pen}_1(|w_j|) = |w_j|$ and $\text{pen}_2(|w_j|) = |w_j| - \text{pen}(|w_j|)$

DC decomposition of the objective function

- Using additivity property of DC
- $J_1(w) = \frac{1}{2} \|y - Xw\|_2^2 + \lambda \sum_{j=1}^d |w_j|$ and $J_2(w) = \lambda \sum_{j=1}^d \text{pen}_2(|w_j|)$

Convex majorization at $w = w_t$

- Majorization of $-J_2(w)$
 $-\lambda \sum_{j=1}^d \text{pen}_2(|w_j|) \leq -\lambda \sum_{j=1}^d \alpha_j^t |w_j| + \text{cte}$ with $\alpha_j^t \in \partial \text{pen}_2(|w_j|)$
- Majorization of the objective function: $J_1(w) - \lambda \sum_{j=1}^d \alpha_j^t |w_j| + \text{cte}$

Iterative re-weighted lasso

Iterative re-weighted Lasso algorithm

Set $t = 0$, initialize w_t

repeat

 Select $\alpha_j^t \in \partial \text{pen}_2(|w_j|)$ for $w = w_t$

 Find $w_{t+1} = \arg \min_w \frac{1}{2} \|y - Xw\|_2^2 + \sum_{j=1}^d (\lambda - \alpha_j^t) |w_j|$

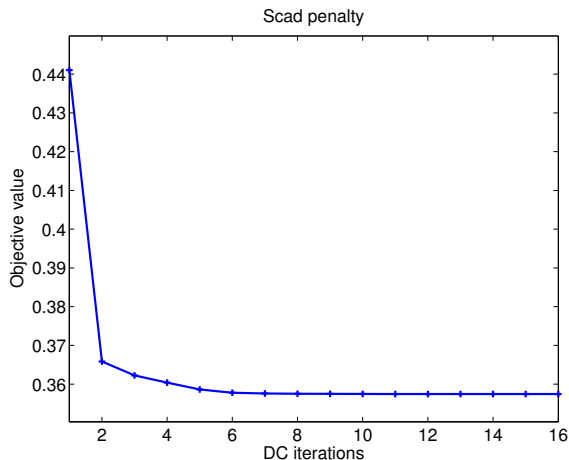
$t = t + 1$

until convergence

- Each iteration is a Lasso type problem
- Require any off-the-shelf Lasso solver

Empirical evaluation: convergence

- Typically few iterations for convergence in objective function



Is there a theoretical guarantee on estimated $\mathbf{w}$?

RIP Condition

Let $\Phi = \begin{pmatrix} \phi(\mathbf{x}_1)^\top \\ \vdots \\ \phi(\mathbf{x}_N)^\top \end{pmatrix} \in \mathbb{R}^{N \times d}$ the dictionary. Φ satisfies the RIP condition at sparsity level $\|\mathbf{w}\|_0 \leq s$ if there exists finite $\underline{c}, \bar{c} > 0$ such that

$$\underline{c}\|\mathbf{w}\|_2^2 \leq \|\Phi\mathbf{w}\|_2^2 \leq \bar{c}\|\mathbf{w}\|_2^2$$

Theorem [ZZ⁺12]

Under RIP condition, previous DC approach for sparse regression gives a solution $\hat{\mathbf{w}}$ with $\text{supp}(\hat{\mathbf{w}}) = \text{supp}(\mathbf{w}^*)$, $\|\hat{\mathbf{w}} - \mathbf{w}^*\|_2^2 \leq O(\sigma^2 \|\mathbf{w}\|_0 / N)$ if for some constant $c > 0$, $\min_{j \in \text{supp}(\mathbf{w}^*)} |w_j^*| \geq c\sigma\sqrt{\ln d / N}$

So far

- Non-convex penalties are effective for support recovery compared to convex penalty
- DC approach promotes multi-stage convex (non-smooth) relaxation to address non-convex (non-smooth) problem
- The convex relaxation may be non-unique
- Prefer decomposition that will lead to "easy" to solve convex problem
- However each iteration requires to solve an entire convex (and possibly computational costly) problem
- How to leverage on fast methods?

Majorization Minimization (MM)

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \mathbf{x} \in \mathcal{X}$$

Majorization Minimization: Iteratively generate a sequence as follows

$$\mathbf{x}_{k+1} \in \arg \min_{\mathbf{x}} \varphi(\mathbf{x}, \mathbf{x}_k) \quad \mathbf{x} \in \mathcal{X}$$

where $\varphi(\mathbf{x}, \mathbf{x}_k)$ is a surrogate function of $f(\mathbf{x})$, satisfying

- $\varphi(\mathbf{x}, \mathbf{x}_k) \geq f(\mathbf{x}) \quad \forall \mathbf{x}_k, \mathbf{x} \in \mathcal{X}$
- $\varphi(\mathbf{y}, \mathbf{y}) = f(\mathbf{y}) \quad \forall \mathbf{y} \in \mathcal{X}$

majorization-minimization scheme (MM)

[Mai14]

Algorithm 4 Basic majorization-minimization scheme (MM)

Data: w initialization

Result: w is a local minima

initialize: find a feasible w

while *not converged* **do**

 Compute a surrogate function s of f near w

 Minimize the surrogate s and update the solution

end

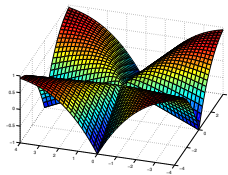
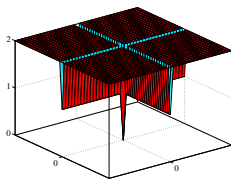
DC and proximal can be seen as MM

Possible issues

- Initial solution, starting point
- Computational efficiency
- When to stop? Local vs. global minima.
- Parameter selection?
- Primal (prox-linear) vs dual (ADM/Bregman)?

Road map

- 1 Parcimonious and non-convex learning problems
- 2 Mixed integer programming for machine learning
- 3 Continuous relaxations
- 4 Nonsmooth and nonconvex optimization tools
- 5 Optimization algorithm for nonconvex & nondifferential functionals
- 6 A case study: linear regression with MCP



A case study: linear regression with MCP

- Objective: compare different non convex optimization methods to provide some recommendations on what to use and when.
- Why MCP? non convex; Minimum concave penalty; Firm shrinkage; nice for implementations;
- Empirical evidence reported confirms
 - ▶ the existence of local minima (and algorithms converging towards it)
 - ▶ the importance of initialization in local optimization procedures.

$$J_{\text{MCP}}(\beta) = \frac{1}{2} \|y - X\beta\|^2 + \sum_{j=1}^p \text{pen}_{\lambda, \gamma}(|\beta_j|),$$

The MCP cost function

$$J_{\text{MCP}}(\beta) = \frac{1}{2} \|y - X\beta\|^2 + \sum_{j=1}^p \text{pen}_{\lambda, \gamma}(|\beta_j|),$$

$$\text{pen}_{\lambda, \gamma}(t) = \lambda \int_0^t \left(1 - \frac{x}{\gamma\lambda}\right)_+ dx = \begin{cases} \lambda t - \frac{t^2}{2\gamma} & \text{if } t \leq \gamma\lambda \\ \frac{\gamma\lambda^2}{2} & \text{else.} \end{cases}$$

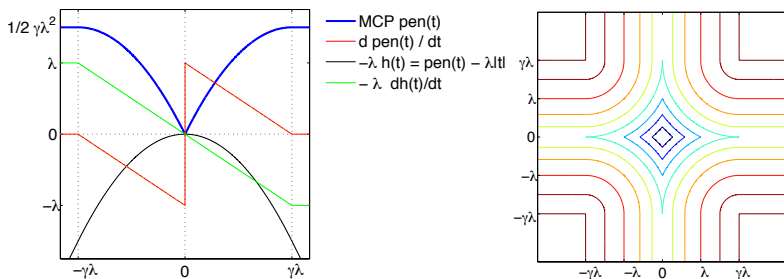


Figure : Left: $\text{pen}_{\lambda, \gamma}(t)$ (blue) for $\lambda = 1$ and $\gamma = 3$, its derivative (red), the associated inverse of the Huber loss (black) and its derivative (green). Right: the MCP penalty level set for $\lambda = \gamma = 2$.

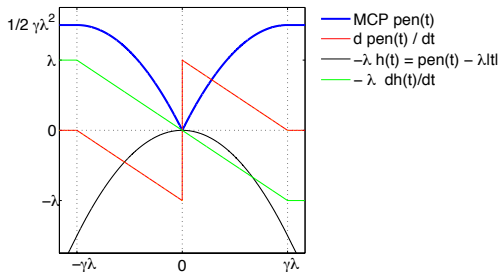
MCP penalty as the difference of two convex functions

$$J_{\text{MCP}}(\beta) = \frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1 - \lambda \sum_{j=1}^p h_{\lambda, \gamma}(|\beta_j|), \quad (3)$$

with

$$h_{\lambda, \gamma}(t) = \left\{ \frac{t^2}{2\gamma\lambda} \mathbb{1}_{\{t \leq \gamma\lambda\}} + \left(t - \frac{\gamma\lambda}{2} \right) \mathbb{1}_{\{t > \gamma\lambda\}} \right\},$$

the Huber penalty function with parameter $\gamma\lambda$,



The MCP cost function as a QP

For all λ there exists a positive scalar k such a stationary point of

$$\left\{ \begin{array}{ll} \min_{\beta \in \mathbb{R}^p} & \frac{1}{2} \|y - X\beta\|^2 - \lambda \sum_{j=1}^p h_{\lambda, \gamma}(|\beta_j|) \\ \text{subject to} & \|\beta\|_1 \leq k \end{array} \right.$$

is also a stationary point of the MCP cost function. The convexity of this reformulation depends on the positiveness of all the Hessian matrix

$$\nabla_{\beta}^2 J_{\text{MCP}}(\beta) = X^{\top} X - \frac{1}{\gamma} \mathbf{diag}(\mathbf{r}),$$

where $\mathbf{r}$ is a vector whose components are 1 if $\beta_j \leq \gamma\lambda$ and 0 else, for all β . MCP is convex when $\gamma > \frac{1}{c^*}$, c^* being the smallest eigen value of $X^{\top} X$.

Table : Example of γ minimum value guaranteeing MCP convexity.

	Housing	Countries	Galaxy	L.A. Ozone	Ozone	Prostate cancer
$\frac{1}{c^*}$	15.75	1.18e+31	17.14	27.05	3.71	6.41

MCP Dini Hadamard subdifferential

The Dini subdifferential of the MCP at some β is the vector set of

$$\partial_{DH} J_{\text{MCP}}(\beta) = X^{\top}(X\beta - y) + \lambda \begin{cases} \alpha_j & \text{if } \beta_j = 0 \\ \text{sgn}(\beta_j) - \frac{\beta_j}{\lambda\gamma} & \text{if } |\beta_j| \leq \lambda\gamma \\ 0 & \text{else} \end{cases} \quad (4)$$

Table : The components of the MCP and their subgradients for $\alpha_j \in [-1, 1]$.

	least square	L_1 penalty	Huber function h
$f(\beta)$	$\frac{1}{2}\ y - X\beta\ ^2$	$\lambda\ \beta\ _1$	$\lambda \sum_{j=1}^p \begin{cases} \frac{\beta_j^2}{2\gamma\lambda} & \text{if } \beta_j \leq \gamma\lambda \\ \beta_j - \frac{\gamma\lambda}{2} & \text{else} \end{cases}$
$\partial_{\beta} f(\beta)$	$X^{\top}(X\beta - y)$	$\lambda \begin{cases} \alpha_j & \text{if } \beta_j = 0 \\ \text{sgn}(\beta_j) & \text{else} \end{cases}$	$\lambda \begin{cases} \frac{\beta_j}{\gamma\lambda} & \text{if } \beta_j \leq \gamma\lambda \\ \text{sgn}(\beta_j) & \text{else} \end{cases}$

MCP critical points (local minima)

for all $\alpha_j \in [-1, 1]$. stationarity conditions are:

$$\left\{ \begin{array}{ll} |g_j| - \lambda \leq 0 & \text{if } \beta_j = 0 \\ g_j + \lambda \operatorname{sgn}(\beta_j) - \frac{\beta_j}{\gamma} = 0 & \text{if } |\beta_j| \leq \lambda\gamma \\ g_j = 0 & \text{else ,} \end{array} \right. \quad (5)$$

with $g = X^\top(X\beta - y)$ the gradient of the square error component of the cost.

Note that when the MCP cost is not convex, it may exist various stationary points and local algorithms are only guaranteed to converge towards a local minimum.

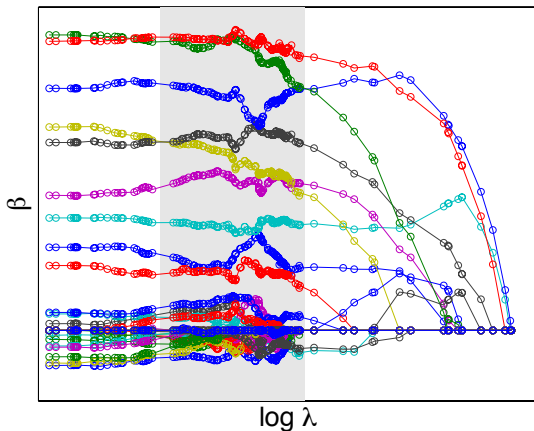
Algorithms for MCP

- the homotopy method (a regularization path algorithm [Zha10]),
- subgradient descent,
- component wise (or shooting, or coordinate descent, or Gauss Seidel [BH11]),
- multi-stage convex relaxation (or DC [GRC09], or LLA, or CCCP, or MM, or reweighed adaptive lasso [ZZ⁺12])
- proximal algorithms
 - ▶ proximal algorithms with MCP proximal [GZL⁺13].
 - ▶ proximal algorithms with L_1 proximal.

The homotopy method for MCP

$$J_{\text{MCP}}(\beta, \lambda) = \frac{1}{2} \|y - X\beta\|^2 + \sum_{j=1}^p \text{pen}_{\lambda, \gamma}(|\beta_j|),$$

$$\beta^*(\lambda) = \arg \min_{\beta} J_{\text{MCP}}(\beta, \lambda)$$



The homotopy method for MCP

Assume that, For a given λ , the three sets I_0, I_P, I_N where respectively $\beta_j = 0, |\beta_j| \leq \gamma\lambda$ and $|\beta_j| > \gamma\lambda$ are given. The non zero components $\beta_I = (\beta_P^\top, \beta_N^\top)^\top$ of vector β , are given by solving

$$\left(X_I^\top X_I - \frac{1}{\gamma} \mathbf{diag}(\mathbf{r}_I) \right) \beta_I(\lambda) = X_I^\top y - \lambda \begin{pmatrix} \text{sign}(\beta_P) \\ 0_N \end{pmatrix}, \quad (6)$$

with $\mathbf{r}_I = \begin{pmatrix} \mathbb{I}_P \\ 0_N \end{pmatrix}$, $\mathbb{I}_P$ being a vector of ones of size $\#I_P$, $I = I_P \cup I_N$ and 0_N being a vector of zeros of size $\#I_N$.

Considering λ' in a neighborhood of λ such that the three sets I_0, I_P, I_N remain unchanged for the solution, equation (6) still hold so that we have

$$\beta_I(\lambda') = \beta_I(\lambda) - (\lambda' - \lambda) \left(X_I^\top X_I - \frac{1}{\gamma} \mathbf{diag}(\mathbf{r}_I) \right)^{-1} \begin{pmatrix} \text{sign}(\beta_P) \\ 0_N \end{pmatrix}.$$

The solution $\beta_I(\lambda)$ varies piecewise linearly with respect to λ .

For the code, see A. Boisbunon's PhD

Subgradient iterations

$$\nabla_{\beta} J_{\text{MCP}}(\beta) = X^{\top}(X\beta - y) + \text{pen}'_{\lambda, \gamma}(|\beta|),$$

When $\beta_j = 0$, any subgradient can be taken. However, when $|g_j| \leq \lambda$, $\beta_j = 0$ fulfill for this component the optimality condition and therefore shouldn't be changed to promote sparsity in the solution.

```
I0 = find(abs(beta)<sqrt(eps));  
g = X'*(X*beta-y);  
g(I0)=(abs(g(I0))-beta(I0)/gam)-lambda.*(abs(g(I0) - beta(I0)/gam) > lambda);  
g = g+((lambda*sign(beta)-beta/gam).*(abs(beta)<lambda*gam)).*(abs(beta)>eps);
```

The component wise algorithm for MCP

$$\frac{\partial J_{\text{MCP}}(\beta)}{\partial \beta_j} = X_{\bullet j}^\top (X\beta - y) + \begin{cases} \lambda \alpha_j & \text{if } |\beta_j| = 0 \\ \text{sign}(\beta_j) \max\left(0, \lambda - \frac{|\beta_j|}{\gamma}\right) & \text{else,} \end{cases}$$

for $\alpha_j \in [-1, 1]$. Vector $X_{\bullet j}$ denotes the column j of matrix X . In that case

$$\beta_j = \begin{cases} 0 & \text{if } |(X_{\bullet j}^\top X_{\bullet j})^{-1} X_{\bullet j}^\top \mathbf{r}| \leq \lambda \\ s_j \left(|(X_{\bullet j}^\top X_{\bullet j})^{-1} X_{\bullet j}^\top \mathbf{r}| - \lambda \right) \frac{\gamma}{\gamma - 1} & \text{if } |(X_{\bullet j}^\top X_{\bullet j})^{-1} X_{\bullet j}^\top \mathbf{r}| \leq \lambda \gamma \\ (X_{\bullet j}^\top X_{\bullet j})^{-1} X_{\bullet j}^\top \mathbf{r} & \text{else,} \end{cases}$$

with s_j the sign of $(X_{\bullet j}^\top X_{\bullet j})^{-1} X_{\bullet j}^\top \mathbf{r}$ and $\mathbf{r} = X\beta - y - X_{\bullet j}\beta_j$ the residual error.

```
for i = 1:nbitemax
    in = randperm(p);
    for j = 1:p
        grad = -(Xi(:,in(j)))'*(Xi*beta-yi-Xi(:,in(j))*beta(in(j))))
        beta(in(j))=sign(grad)*max(0,min((abs(grad)-lam)/(1-1/gam),abs(grad)));
    end
end
```

The proximal of the MCP penalty

$$\mathbf{prox}_{\text{MCP}}(u) = \arg \min_{t \in \mathbb{R}} \frac{1}{2} \|t - u\|^2 + \mathbf{pen}_{\lambda, \gamma}(|t|) .$$

The solution of this minimization problem gives

$$\mathbf{prox}_{\text{MCP}}(u) = \begin{cases} 0 & \text{if } |u| \leq \lambda \\ \text{sign}(u) (|u| - \lambda) \frac{\gamma}{\gamma - 1} & \text{if } \lambda < |u| \leq \lambda\gamma \\ u & \text{else.} \end{cases}$$

$$\beta^{\text{new}} = \mathbf{prox}_{\text{MCP}}(\beta^{\text{old}} - \rho X^{\top}(X\beta - y)),$$

where $\rho > 0$ is a step size, constant or determined by line search.

```
rho = 1/svds(Xi,1)^2;
while max(abs(beta-b0)) > tol
    b0 = beta;
    g = X'*(X*beta-y);
    bg = beta - rho * g;
    s = min(abs(bg), (abs(bg)-lam*rho)*gam/(gam-pas));
    beta = sign(bg).*max(0,s);
end
```

Using the proximal of the L_1 norm

$\phi(\beta) = \frac{1}{2}\|y - X\beta\|_2^2 - h(\beta)$ and a non differentiable one $\varphi(\beta) = \lambda\|\beta\|_1$.

$$\nabla_{\beta}\phi(\beta) = X^{\top}(X\beta - y) - \begin{cases} \text{sign}(\beta_j)\lambda & \text{if } |\beta_j| > \lambda\gamma \\ \frac{\beta_j}{\gamma} & \text{else.} \end{cases}$$

The proximal operator of the L_1 penalty function φ is

$$\text{prox}_{\varphi}(u) = \begin{cases} 0 & \text{if } |u| \leq \lambda \\ \text{sign}(u)(|u| - \lambda) & \text{else.} \end{cases}$$

```
while max(abs(beta-b0)) > tol
    b0 = beta;
    g = Xi'*(Xi*beta-yi) - min(lam,abs(beta)/gam).*sign(beta);
    bg = beta - rho * g;
    beta = sign(bg).*max(0,abs(bg) - rho*lam);
end
```

The DC algorithm for MCP

$$J_{\text{DC}}(\beta) = \frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1 - \lambda \sum_{j=1}^p h'_{\lambda, \gamma}(|\beta_j^{\text{old}}|) |\beta_j|, \quad (7)$$

h' being the derivative of the Huber loss function. minimizing the DC cost function (7) can be seen as minimizing an adaptive Lasso [Zou06]

$$\min_{\beta \in \mathbb{R}^p} \frac{1}{2} \|y - X\beta\|^2 + \lambda \sum_{j=1}^p w_j |\beta_j|, \quad w_j = \begin{cases} 0 & \text{if } |\beta_j^{\text{old}}| > \lambda\gamma \\ 1 - \frac{|\beta_j^{\text{old}}|}{\gamma\lambda} & \text{else} \end{cases}$$

```
while max(abs(beta-b0)) > tol
    b0 = beta;
    w = max(0, lambda - abs(beta)/gam);
    [xnew, dual_var, pos] = monqp(H,c,[w ; w],b,inf,1,0);
    Bpm = zeros(2*p,1);
    Bpm(pos) = xnew;
    beta = Bpm(1:p)-Bpm(p+1:end);
end
```

Experimental setup

X be an $n \times p$ design matrix and $y \in \mathbb{R}^n$ a response vector

$$Y = X\beta + \varepsilon, \quad \beta \in \mathbb{R}^p. \quad (8)$$

- X has been also randomly generated using a correlated Gaussian normalized distribution whose vector components were correlated with correlation between row j and k of matrix X equals r^{k-j} [Zou06]
- As usual, the design matrix X has been normalized column wise to mean zero and variance one.
- For all $p \geq 10$ the true value of the parameter was $\beta = (1, 2, 3, 4, 5, -1, -2, -3, -4, -5, 0, \dots, 0)^\top$ with ten non zero components completed with zeros.
- All methods were run until the same convergence criteria was met (*i.e.* where appropriate, that the step length between iterates, change in function value between iterates, negative directional derivative, or optimality condition was less than 10^{-6}).

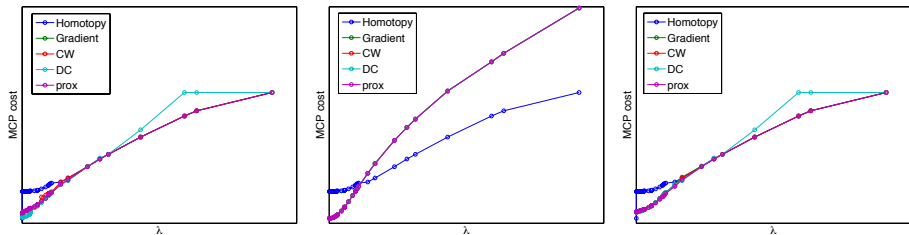


Figure : Illustration of the influence of the initialization. The MCP costs of the whole regularization paths of seven algorithms for $n = 20, p = 15, \gamma = 2.5$ (non convex case) when starting from zero (left), the least square solution (center) and the Lasso (right).

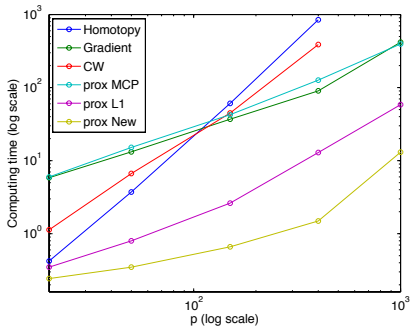
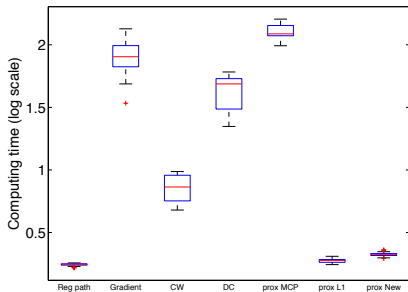


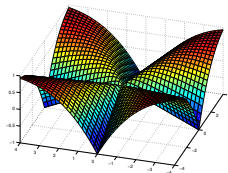
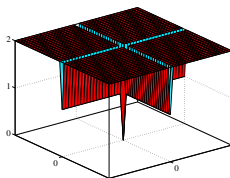
Figure : Left: the computing times of twenty replications of the whole regularization paths of seven algorithms for $n = 20000$, $p = 20$, $\gamma = 2$. Right: the computing time complexity curves as a function of p for $n = 20000$, $\gamma = 2$.

Experimental results

- The convex case
 - ▶ DC may present unstable behavior depending on the solver used for the adaptive lasso.
 - ▶ Convergence criteria is an issue
- The non convex case
 - ▶ There may exist many stationary points. [BH11] note that the solutions are continuous and stable in the locally convex regions, but may be discontinuous and erratic otherwise.
 - ▶ The solution provided by most of the algorithms strongly depends on the initial guess. Begin with low cost (least square with $\lambda = 0$), with a sparse solution ($\beta = 0$ and $\lambda = \infty$) or with the solution provided by the Lasso as advocated in [WLZ14].
 - ▶ The homotopy method provides a different solution. It may take a sub optimal path (only in the non convex case)
- Large scale
 - ▶ Homotopy scales well with n
 - ▶ Homotopy hardly scale with p
 - ▶ For small λ prox's is two to ten times faster than CW.

Road map

- 1 Parcimonious and non-convex learning problems
- 2 Mixed integer programming for machine learning
- 3 Continuous relaxations
- 4 Nonsmooth and nonconvex optimization tools
- 5 Optimization algorithm for nonconvex & nondifferential functionals
- 6 A case study: linear regression with MCP



Conclusion

- Nonconvex Nondifferentiable nature of machine learning
- Consider Mixed integer programming
 - ▶ more improvements to come
- Continuous Nonconvex Nondifferentiable relaxation
 - ▶ Optimization theory to be specialized
 - ▶ Flourishing efficient optimization algorithms
 - ▶ Appear effective in practice to aggressively enforce sparsity
 - ▶ Many extensions for classification, regression, matrix factorization
 - ▶ Extension to case where one seeks sparsity in the loss function side (example : SVM)
 - ▶ Extension to structured sparsity
 - ▶ Lack of theoretical analysis of local optimal solution

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